

International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Firewall Traffic Control and Forensic Verification

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EXECUTIVE SUMMARY

This assignment presents a practical digital forensics investigation focused on the collection, preservation, examination, and interpretation of DNS network traffic. The investigation was conducted within a controlled Kali Linux environment using DNS queries generated against example.com and packet-level evidence captured using TShark. The exercise began with preparation and validation of the forensic workspace, followed by examination of the system's DNS resolver configuration, active network interface, and available packet-capture interfaces. Baseline DNS queries were performed for A, AAAA, MX, and NS records to establish reference results before network traffic was captured.

The investigation also involved the acquisition and preservation of a supplied DNS packet capture, creation of a separate working copy, and SHA-256 integrity verification. A fresh DNS capture was subsequently generated and preserved as PCAPNG evidence. Detailed packet analysis identified six DNS packets representing three complete query-and-response transactions for A, AAAA, and MX records. The transactions were examined using DNS transaction identifiers, source and destination addresses, UDP ports, response codes, answer counts, returned records, TTL values, DNS flags, recursion information, and EDNS fields.

The captured A and AAAA responses corresponded with the previously established baseline results, while the MX response returned the same preference and root-label

mail-exchange configuration observed during baseline testing. All captured responses returned RCODE 0 (NOERROR), and each request was successfully matched to its corresponding response. The captured traffic was low-volume and limited to a single domain and resolver, with no repeated queries, alternate DNS destinations, DNS resolution failures, or clear indicators of DNS tunnelling observed within the available evidence. The investigation demonstrates the practical application of basic digital-forensics principles to DNS traffic, including evidence preservation, integrity verification, controlled acquisition, packet analysis, baseline comparison, and evidence-based reporting.

PURPOSE OF THE ASSIGNMENT

The purpose of this assignment is to develop practical skills in the collection, preservation, examination, and forensic interpretation of DNS network traffic. The exercise provides an opportunity to apply basic computer and digital-forensics techniques to a controlled network environment by establishing a DNS baseline, capturing DNS communications, examining packet-level evidence, validating evidence integrity, and documenting observable findings. It is also intended to demonstrate how commonly available command-line and network-analysis tools can be used to investigate DNS requests and responses, understand resolver behavior, correlate network transactions, and identify potential indicators of abnormal DNS activity.

OBJECTIVES OF THE ASSIGNMENT

The objectives of the assignment are to:

- Prepare and validate a structured digital-forensics workspace.
- Examine and document the system's DNS resolver configuration and active network interface.
- Verify the availability and versions of the tools required for DNS and packet analysis.
- Establish a DNS baseline by performing A, AAAA, MX, and NS record queries.
- Acquire and preserve DNS packet-capture evidence.
- Create and verify a working copy of captured evidence without altering the original evidence.

- Calculate SHA-256 hashes to establish and verify evidence integrity.
- Capture controlled DNS traffic using TShark.
- Identify and correlate DNS query-and-response transactions using transaction identifiers.
- Examine DNS packet fields, including source and destination addresses, ports, query types, response codes, answer records, TTL values, and DNS flags.
- Analyse EDNS information contained within the captured DNS packets.
- Compare captured DNS responses with established baseline results.
- Examine DNS communication behaviour for repeated queries, unusual destinations, failed resolutions, or other observable indicators.
- Assess the available evidence for potential DNS tunnelling or other abnormal DNS characteristics.
- Document the investigation using reproducible commands, preserved outputs, screenshots, tables, and forensic observations.
- Produce an evidence-based conclusion that clearly states the scope and limitations of the captured data.

METHODOLOGY OF THE ASSIGNMENT

The assignment was conducted using a structured digital-forensics methodology covering preparation, baseline establishment, evidence acquisition, preservation, examination, analysis, correlation, and reporting. The investigation began by preparing a dedicated forensic workspace and verifying the operating environment, DNS resolver configuration, active network interface, packet-capture interfaces, and required analysis tools. Baseline DNS information was then collected using dig by querying A, AAAA, MX, and NS records for example.com. A supplied DNS packet capture was acquired and preserved in the evidence directory, after which a separate working copy was created and SHA-256 hashing was used to verify that the working copy remained identical to the original evidence. Controlled DNS queries were subsequently generated while TShark monitored DNS traffic on the active interface. The resulting PCAPNG capture was validated and preserved as forensic evidence. TShark was then used to extract and analyse DNS transaction identifiers, source and destination addresses, UDP ports, query names, query types, response codes, answer counts, returned records, TTL values, DNS flags,

recursion information, and EDNS fields. Individual packets were examined to establish the relationship between DNS requests and responses, after which the captured results were compared with the previously established baseline. The investigation also assessed the DNS traffic for observable indicators such as repeated queries, high-volume activity, unusual query names, multiple DNS destinations, DNS resolution failures, and potential tunnelling characteristics. Evidence integrity was maintained through hashing and preservation of the relevant files, while analysis results were recorded in structured report files. The final findings were interpreted strictly within the scope of the available packet evidence and the short capture period.

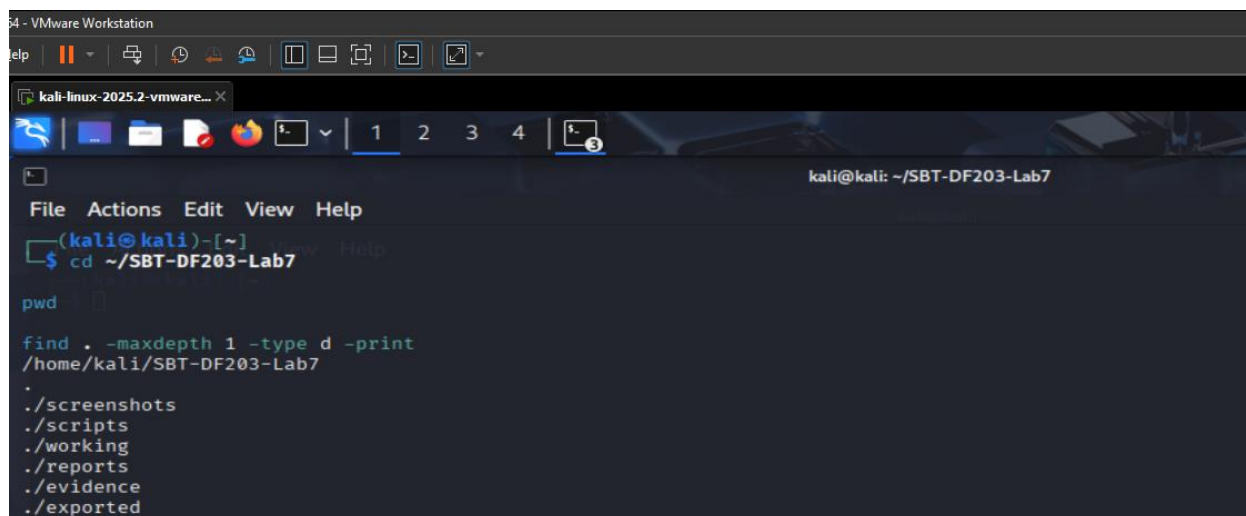
TOOLS AND RESOURCES USED

Tool / Resource	Purpose
Kali Linux	Primary operating system and forensic analysis environment.
Bash / Linux Terminal	Execution of forensic, DNS, file-management, hashing, and analysis commands.
dig	DNS queries and establishment of A, AAAA, MX, and NS baseline results.
TShark	DNS packet capture, filtering, field extraction, transaction correlation, and analysis.
Wireshark	Packet-level and protocol-level examination of DNS traffic.
capinfos	Examination of PCAP/PCAPNG metadata, packet count, file size, duration, and format.
sha256sum	SHA-256 evidence integrity verification.
resolvectl	Examination of active DNS servers, resolver configuration, and network interface information.
/etc/resolv.conf	Documentation of the local DNS stub resolver configuration.
wget	Acquisition of the supplied DNS packet capture.

Tool / Resource	Purpose
cp --preserve=timestamps	Creation of a working copy while preserving available file timestamps.
find / ls / pwd	Workspace, file, directory, and evidence verification.
Git / GitHub	Version control and preservation of laboratory documentation, analysis outputs, and supporting files.
DNS PCAP/PCAPNG Evidence	Primary packet-level evidence examined during the investigation.
example.com	Controlled domain used to generate DNS queries for analysis.
8.8.8.8	External recursive DNS resolver observed in the captured traffic.
DNS / EDNS Protocol Fields	Protocol information examined during packet-level forensic analysis.

Lab Workspace Preparation

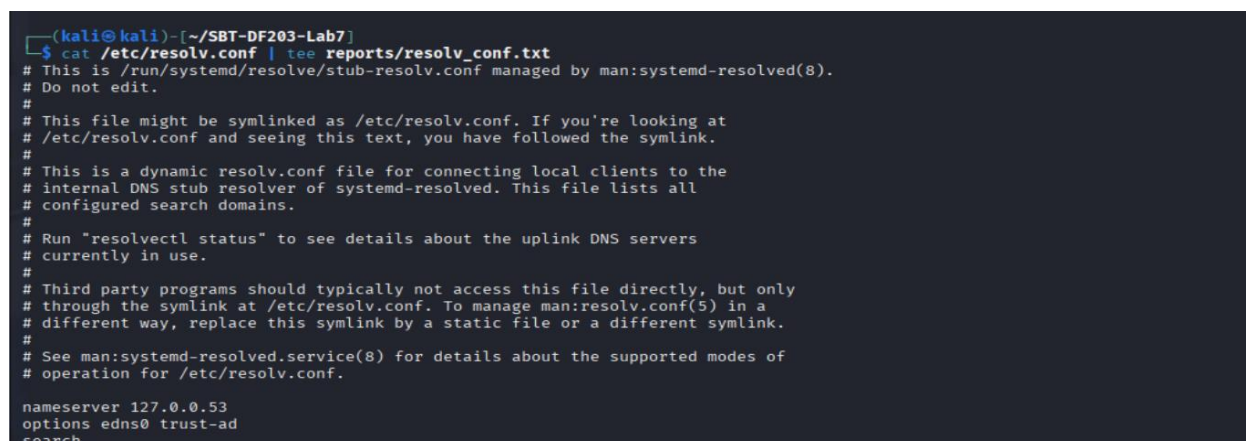
The DNS forensic analysis workspace was prepared under /home/kali/SBT-DF203-Lab7. The required directories were created to provide a structured location for evidence collection, working files, exported analysis results, reports, screenshots, and supporting scripts. The workspace structure was subsequently verified using pwd and find, confirming that the expected directories (evidence, working, exported, reports, screenshots, and scripts) were available and ready for use throughout the investigation.



```
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
(kali@kali)-[~]
$ cd ~/SBT-DF203-Lab7
pwd
/home/kali/SBT-DF203-Lab7
find . -maxdepth 1 -type d -print
./
./screenshots
./scripts
./working
./reports
./evidence
./exported
```

DNS Resolver Configuration

The system DNS configuration was examined by reviewing `/etc/resolv.conf`. The output showed that the system is using the `systemd-resolved` stub resolver, with `127.0.0.53` configured as the local nameserver. The configuration also indicates that the file is dynamically managed by `systemd-resolved` and should not be manually edited. A copy of the configuration was saved as `reports/resolv_conf.txt` to preserve the initial DNS resolver state as part of the forensic baseline.



```
(kali@kali)-[~/SBT-DF203-Lab7]
$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
nameserver 127.0.0.53
options edns0 trust-ad
search .
```

Upstream DNS and Network Interface Verification

The active DNS configuration was further examined using `resolvectl status`. The results confirmed that `eth0` is the active default-route interface with DNS functionality enabled. The configured upstream DNS servers were identified as `1.1.1.1` and `8.8.8.8`, with `8.8.8.8` reported as the current DNS server at the time of collection. DNSSEC was reported as unsupported or disabled, while LLMNR was enabled on the relevant interface. The

resolvectl output was saved to reports/resolvectl_status.txt to preserve the system's DNS and network resolver state before further analysis.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
Global
  Protocols: +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 3 (docker0)
  Current Scopes: none
  Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Default Route: no

Link 4 (br-d2696585dc9a)
  Current Scopes: LLMNR/IPv4 LLMNR/IPv6
  Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Default Route: no

Link 2 (eth0)
  Current Scopes: DNS LLMNR/IPv4 LLMNR/IPv6
  Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Current DNS Server: 8.8.8.8
  DNS Servers: 1.1.1.1 8.8.8.8
  Default Route: yes

Link 5 (br-ec90d8c3abfb)
  Current Scopes: none
  Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Default Route: no

Link 9 (veth8a933b9)
  Current Scopes: none
  Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Default Route: no

Link 6 (veth356d7eb)
  Current Scopes: LLMNR/IPv6
  Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Default Route: no
```

Network Capture Interface Verification

The available packet-capture interfaces were enumerated using TShark with the `tshark -D` command. Multiple interfaces were detected, including `eth0`, loopback, Docker bridges, virtual Ethernet interfaces, and other capture mechanisms. The primary network interface identified for the host was `eth0`, which corresponds with the interface reported by `resolvectl` as the active default-route and DNS-capable interface. This verification establishes the appropriate interface for subsequent DNS traffic capture.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -D
Warning: program compiled against libxml 215 using older 214
1. eth0
2. br-d2696585dc9a
3. veth356d7eb
4. veth8a933b9
5. any
6. lo (Loopback)
7. docker0
8. br-ec90d8c3abfb
9. bluetooth-monitor
10. nflog
11. nfqueue
12. dbus-system
13. dbus-session
14. ciscodump (Cisco remote capture)
15. dpauemon (DisplayPort AUX channel monitor capture)
16. randpkt (Random packet generator)
17. sdjournal (systemd Journal Export)
18. sshdig (SSH remote syscall capture)
19. sshdump (SSH remote capture)
20. udpdump (UDP Listener remote capture)
21. wifidump (Wi-Fi remote capture)
```

Tool and Version Verification

The availability and versions of the primary DNS and packet-analysis tools were verified before commencing the forensic analysis. dig was confirmed to be installed as version 9.20.27-2-Debian, while TShark and Wireshark were confirmed as version 4.6.6. The system was running Kali Linux with kernel version 6.12.33+kali-amd64. This verification establishes that the required DNS diagnostic and packet-analysis utilities are available and provides a record of the software environment used during the investigation.

```

kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig -v
tshark --version
wireshark --version
Dig 9.20.27-2-Debian
Warning: program compiled against libxml 215 using older 214
TShark (Wireshark) 4.6.6.

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This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
Bit width: 64-bit
Compiler: GCC 15.2.0
GLib: 2.88.1

With:
+ Brotli
+ Gcrypt 1.12.2
+ GnuTLS 3.8.13 and PKCS#11
+ Kerberos (MIT)
+ libnl 3
+ libpcap
+ libsmi 0.4.8
+ libxml2 2.15.2
+ lua 5.4.8
+ LZ4 1.10.0
+ MaxMind
+ nghttp2 1.69.0
+ nghttp3 1.15.0
+ PCRE2 10.46 2025-08-27
+ POSIX capabilities (Linux)
+ Snappy 1.2.2
+ xxhash 0.8.3
+ zlib 1.3.2
+ Zstandard 1.5.7

Without:
- zlib-ng

Runtime info:
OS: Linux 6.12.33+kali-amd64
CPU: Intel(R) Core(TM) i7-4600U CPU @ 2.10GHz (with SSE4.2)
Memory: 6894 MB of physical memory
GLib: 2.84.3
Locale: LC_TYPE=en_US.UTF-8
Plugins: supported, 0 loaded

With:
+ Brotli 1.1.0
+ c-ares 1.34.5
+ Gcrypt 1.12.2
+ GnuTLS 3.8.9
+ nghttp2 1.64.0
+ nghttp3 1.17.0
+ PCRE2 10.45 2025-02-05
+ xxhash 803

```

```
nd64 - VMware Workstation
Help [Icons]

kali-linux-2025.2-vmware... X
[Icons] | 1 2 3 4 | 5

kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help

+LZ4 1.10.0
Wireshark 4.6.6.

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This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
  Bit width: 64-bit
  Compiler: GCC 15.2.0
  GLib: 2.88.1
With:
  +brotli
  +Gcrypt 1.12.2
  +GnuTLS 3.8.13 and PKCS#11
  +Kerberos (MIT)
  +libnl 3
  +libpcap
  +libsmi 0.4.8
  +libxml2 2.15.2
  +Lua 5.4.8
  +LZ4 1.10.0
  +MaxMind
  +Minizip 1.3.2
  +nghttp2 1.69.0
  +nghttp3 1.15.0
  +PCRE2 10.46 2025-08-27
  +POSIX capabilities (Linux)
  +Qt 6.10.2
  +QtDBus
  +QtMultimedia
  +Snappy 1.2.2
  +xxhash 0.8.3
  +zlib 1.3.2
  +Zstandard 1.5.7
Without:
  -automatic updates -zlib-ng

Runtime info:
  OS: Linux 6.12.33+kali-amd64
  CPU: Intel(R) Core(TM) i7-4600U CPU @ 2.10GHz (with SSE4.2)
  Memory: 6894 MB of physical memory
  GLib: 2.84.3
  Locale: LC_TYPE=en_US.UTF-8
  Plugins: supported, 0 loaded
With:
  +brotli 1.1.0
  +c-ares 1.34.5
  +Gcrypt 1.12.2
  +GnuTLS 3.8.9
  +libpcap 1.10.5 (with TPACKET_V3)
  +libsmi 0.4.8
  +LZ4 1.10.0
  +nghttp2 1.64.0
  +nghttp3 1.17.0
  +PCRE2 10.45 2025-02-05
  +Qt 6.10.2
  +xxhash 803
  +zlib 1.3.1
  +Zstandard 1.5.7
```

At this stage of the investigation, the DNS forensic analysis environment has been successfully prepared and validated. The workspace structure was created, the local DNS resolver configuration was documented, the upstream DNS servers and active network interface were identified, and the available packet-capture interfaces were verified. The required DNS and network-analysis tools were also confirmed to be installed and operational. These steps establish the initial system baseline required before generating and capturing DNS traffic for subsequent forensic analysis.

A Record Baseline and DNS Query

A baseline DNS A-record query was performed for example.com using the dig utility. The query completed successfully with a NOERROR status and returned two IPv4 addresses:

104.20.23.154 and 172.66.147.243, each with a 300-second TTL. The query was resolved through the local systemd-resolved stub at 127.0.0.53 over UDP port 53. The result was saved to reports/dig_example_A.txt for subsequent reference and comparison during the forensic analysis.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com A | tee reports/dig_example_A.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 11584
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 300     IN      A      104.20.23.154
example.com.                 300     IN      A      172.66.147.243

;; Query time: 271 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:07:33 EDT 2026
;; MSG SIZE rcvd: 72
```

AAAA Record Baseline and DNS Query

An AAAA-record query was performed for example.com to establish the domain's IPv6 DNS response. The query returned successfully with a NOERROR status and produced two IPv6 addresses: 2606:4700:10::6814:179a and 2606:4700:10::ac42:93f3, both with a 300-second TTL. The query was processed through the local DNS stub resolver at 127.0.0.53. The complete response was preserved in reports/dig_example_AAAA.txt.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 24937
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                 300     IN      AAAA    2606:4700:10::6814:179a
example.com.                 300     IN      AAAA    2606:4700:10::ac42:93f3

;; Query time: 331 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:07:55 EDT 2026
;; MSG SIZE rcvd: 96
```

MX Record Baseline and DNS Query

An MX-record query was performed against example.com to establish its mail-exchange configuration. The DNS server returned a successful NOERROR response containing one MX record, with a preference value of 0 pointing to the root label (.). This indicates that the queried domain does not designate a conventional mail-exchange host for email delivery. The response was obtained through the local DNS stub resolver and saved to reports/dig_example_MX.txt for forensic documentation.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com MX | tee reports/dig_example_MX.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 35428
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                286     IN      MX      0 .

;; Query time: 88 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:08:05 EDT 2026
;; MSG SIZE rcvd: 55
```

NS Record Baseline and DNS Query

An NS-record query was performed for example.com to identify the authoritative name servers associated with the domain. The query returned successfully and identified hera.ns.cloudflare.com and elliot.ns.cloudflare.com as the two name servers returned in the response, with a TTL of 21,600 seconds. The result was resolved through the local DNS stub at 127.0.0.53 and saved to reports/dig_example_NS.txt.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com NS | tee reports/dig_example_NS.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 63344
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                21600   IN      NS      hera.ns.cloudflare.com.
example.com.                21600   IN      NS      elliot.ns.cloudflare.com.

;; Query time: 84 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:08:16 EDT 2026
;; MSG SIZE rcvd: 95
```

Short DNS Response Check and Resolution Check

A concise DNS lookup was also performed using `dig +short example.com A`. This command returned only the resolved IPv4 addresses without the additional DNS response metadata, producing 172.66.147.243 and 104.20.23.154. This provided a simplified confirmation of the A-record resolution and was saved to `reports/dig_example_short.txt`.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig +short example.com A | tee reports/dig_example_short.txt
172.66.147.243
104.20.23.154

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

A normal DNS query baseline was established for `example.com` by querying its A, AAAA, MX, and NS record types. The queries completed successfully with NOERROR responses and provided baseline information covering IPv4 and IPv6 resolution, mail-exchange configuration, and authoritative name servers. A shortened A-record lookup was also performed to confirm the resolved IPv4 addresses. All query outputs were saved under the `reports` directory, providing preserved reference data for comparison with DNS traffic and responses examined during subsequent forensic analysis.

DNS Packet Capture and PCAP Acquisition

A DNS packet capture file was obtained from the supplied Digital Forensics Lab repository using `wget`. The file was successfully downloaded and saved as `evidence/dig_dns.pcap` within the evidence directory. The download completed with an HTTP 200 OK response, confirming successful retrieval of the 1.72 MB packet capture. The original capture was retained in the evidence directory as the source evidence for subsequent DNS traffic analysis.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
--2026-09-21 15:12:19-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1804656 (1.7M) [application/octet-stream]
Saving to: 'evidence/dig_dns.pcap'

evidence/dig_dns.pcap      100%[=====>] 1.72M  3.19MB/s  in 0.5s

2026-09-21 15:12:21 (3.19 MB/s) - 'evidence/dig_dns.pcap' saved [1804656/1804656]
```

Working Copy Creation

To preserve the original evidence file, a working copy of the DNS packet capture was created using `cp --preserve=timestamps`. The original file, `evidence/dig_dns.pcap`, was copied to `working/dig_dns_working.pcap`. Preserving the timestamps helps maintain the available file metadata while allowing subsequent analysis to be performed against a separate working copy rather than directly modifying the original evidence.

```
(kali@kali)~[~/SBT-DF203-Lab7]
$ wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
--2026-09-21 15:12:19-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1804656 (1.7M) [application/octet-stream]
Saving to: 'evidence/dig_dns.pcap'

evidence/dig_dns.pcap      100%[=====] 1.72M  3.19MB/s  in 0.5s
2026-09-21 15:12:21 (3.19 MB/s) - 'evidence/dig_dns.pcap' saved [1804656/1804656]

(kali@kali)~[~/SBT-DF203-Lab7]
$ cp --preserve=timestamps \
evidence/dig_dns.pcap \
working/dig_dns_working.pcap
```

SHA-256 DNS Capture Integrity Verification

SHA-256 hashes were calculated for both the original evidence capture and the working copy to establish and verify file integrity. Both files produced the identical SHA-256 hash, `9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d`. The matching hashes confirm that the working copy is byte-for-byte consistent with the original evidence file at the time of verification. The hash values were recorded in `reports/dns_capture_hashes.txt` to provide an auditable integrity record.

```
(kali@kali)~[~/SBT-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap \
| tee reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d working/dig_dns_working.pcap

(kali@kali)~[~/SBT-DF203-Lab7]
$ cat reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d working/dig_dns_working.pcap
```

Evidence Handling Summary

The supplied DNS packet capture was acquired and placed in the designated evidence directory before analysis. A separate working copy was created to support forensic examination while preserving the original evidence. SHA-256 hashing was subsequently performed on both files, and the identical hash values confirmed that no alteration occurred during the copying process. This establishes an initial chain-of-custody and integrity baseline for the DNS packet capture prior to further examination.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap \
| tee reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  working/dig_dns_working.pcap

(kali@kali)-[~/SBT-DF203-Lab7]
$ cat reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  working/dig_dns_working.pcap

```

Live DNS Capture

Following the earlier capture setup issue, the DNS traffic capture was successfully initiated using TShark on the eth0 interface. A capture filter for port 53 was applied to monitor DNS traffic, and the capture was configured for a 25-second period. To avoid the earlier output-file permission issue, the capture was temporarily written to /tmp/fresh_dig_dns.pcapng. TShark reported that the capture was active on eth0 and completed with six captured packets, providing a fresh packet-level dataset for subsequent analysis.

```

kali@kali: ~/SBT-DF203-Lab7
ls -ld ~/SBT-DF203-Lab7 ~/SBT-DF203-Lab7/evidence
drwxrwxr-x 2 kali kali 4096 Sep 21 15:12 evidence
drwxrwxr-x 8 kali kali 4096 Sep 21 14:47 /home/kali/SBT-DF203-Lab7
drwxrwxr-x 2 kali kali 4096 Sep 21 15:12 /home/kali/SBT-DF203-Lab7/evidence

(kali@kali)-[~/SBT-DF203-Lab7]
$ ls -la evidence/fresh_dig_dns.pcapng 2>/dev/null || true

(kali@kali)-[~/SBT-DF203-Lab7]
$ ps aux | grep 'tshark'

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 \
-w evidence/fresh_dig_dns.pcapng 0
[1] 46403

(kali@kali)-[~/SBT-DF203-Lab7]
$ Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
tshark: The file to which the capture would be saved ("evidence/fresh_dig_dns.pcapng") could not be opened: Permission denied.

[1] + exit 1      sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
(kali@kali)-[~/SBT-DF203-Lab7]
$ fg
fg: no current job

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w /tmp/fresh_dig_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'

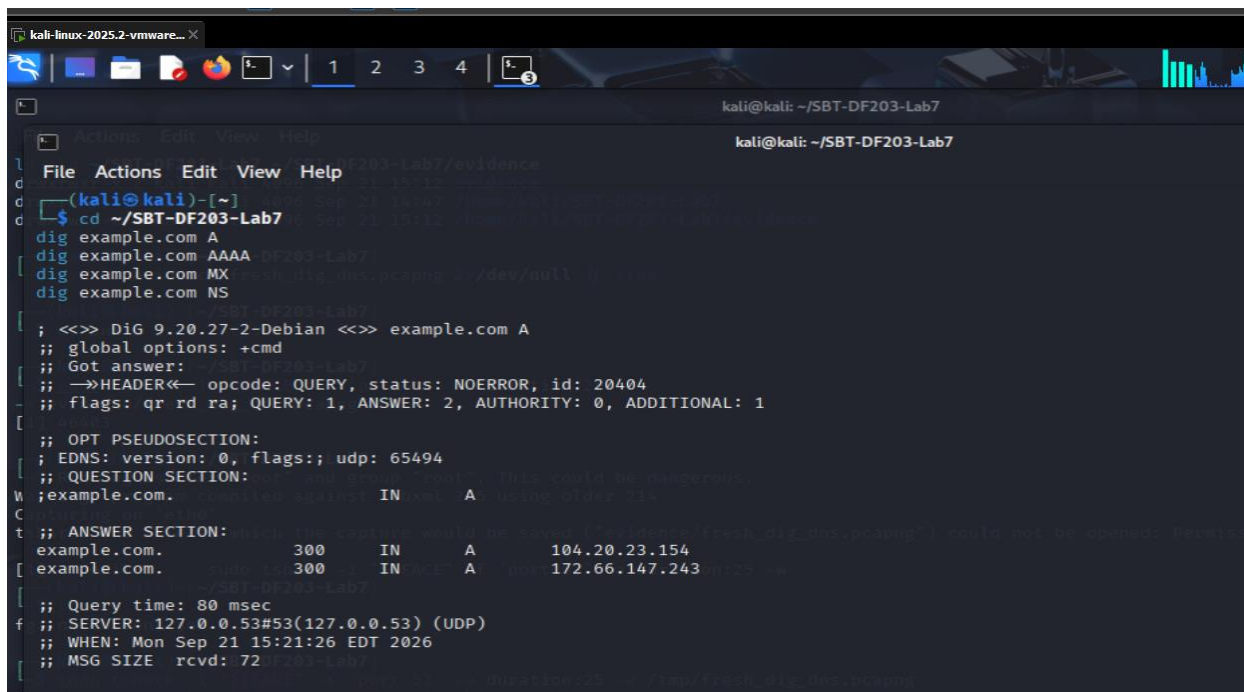
(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w /tmp/fresh_dig_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
6

```

Live A Record Query

During the DNS traffic capture, an A-record query was performed for example.com using dig. The query returned a NOERROR response through the local DNS stub resolver at

127.0.0.53 and returned two IPv4 addresses: 104.20.23.154 and 172.66.147.243, each with a 300-second TTL. This query generated DNS request and response traffic that could be examined in the packet capture.



```
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
(kali@kali)-[~]
$ cd ~/SBT-DF203-Lab7
dig example.com A
dig example.com AAAA
dig example.com MX
dig example.com NS

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 20404
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com.                300     IN      A      104.20.23.154
example.com.                300     IN      A      172.66.147.243

;; Query time: 80 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:21:26 EDT 2026
;; MSG SIZE rcvd: 72
```

Live AAAA Record Query

An AAAA-record query was subsequently performed for example.com to obtain its IPv6 resolution information. The query completed successfully with a NOERROR response and returned two IPv6 addresses, 2606:4700:10::ac42:93f3 and 2606:4700:10::6814:179a, with a TTL of 300 seconds. The response was obtained through the local DNS resolver at 127.0.0.53, providing additional DNS traffic for comparison during packet-level analysis.



```
kali@kali: ~/SBT-DF203-Lab7
; <<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 24472
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      AAAA
;; ANSWER SECTION:
example.com.                300     IN      AAAA   2606:4700:10::ac42:93f3
example.com.                300     IN      AAAA   2606:4700:10::6814:179a

;; Query time: 84 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:21:27 EDT 2026
;; MSG SIZE rcvd: 96
```

Live MX Record Query

An MX-record query was performed for example.com as part of the controlled DNS query sequence. The query returned a successful NOERROR response containing an MX record with a preference value of 0 and a target of .. The response was obtained through the local DNS stub resolver at 127.0.0.53. This query was included in the capture sequence to provide a baseline for examining different DNS record types in the packet data.

```
; <<>> DiG 9.20.27-2-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 20375
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      MX
;

;; ANSWER SECTION:
example.com.                55      IN      MX      0 .
;

;; Query time: 104 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:21:27 EDT 2026
;; MSG SIZE rcvd: 55
```

Live NS Record Query

Finally, an NS-record query was performed for example.com to identify the name servers returned for the domain. The query completed successfully with a NOERROR response and returned `elliott.ns.cloudflare.com` and `hera.ns.cloudflare.com`, each identified as an NS record. The response was resolved through the local DNS stub at 127.0.0.53. This completed the controlled DNS query sequence used to generate traffic for subsequent packet analysis.

```
; <<>> DiG 9.20.27-2-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 45902
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

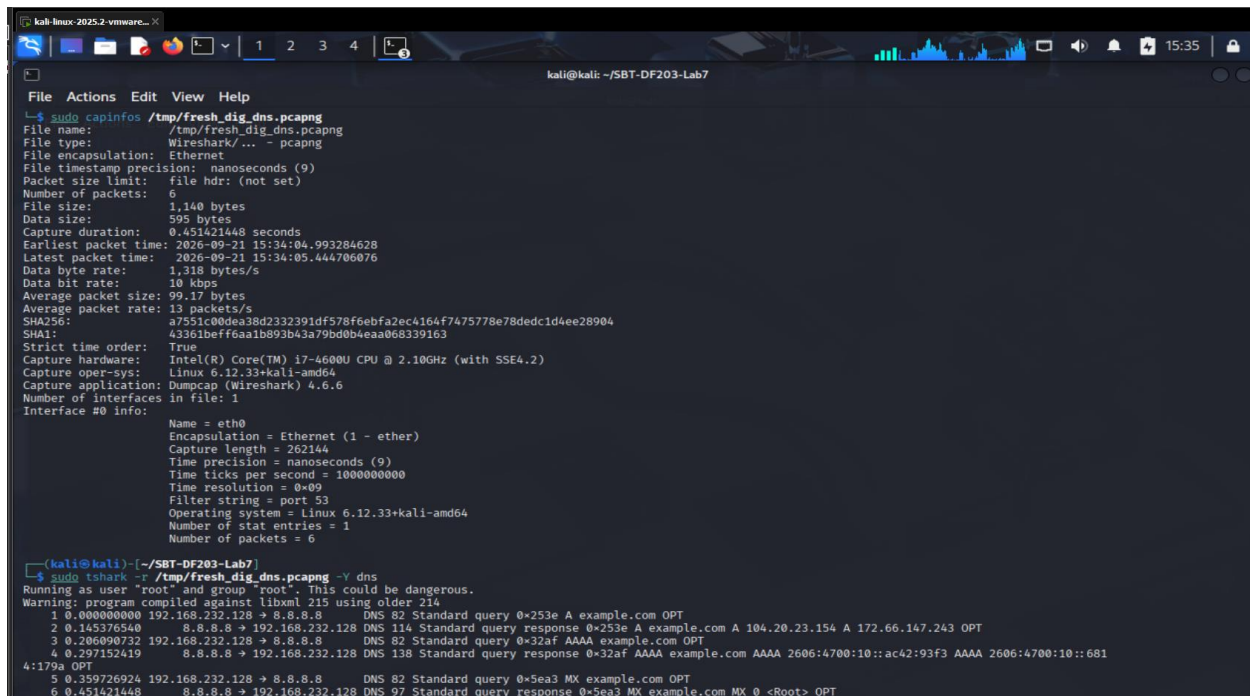
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      NS
;

;; ANSWER SECTION:
example.com.                6408    IN      NS      elliott.ns.cloudflare.com.
example.com.                6408    IN      NS      hera.ns.cloudflare.com.
;

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 15:21:27 EDT 2026
;; MSG SIZE rcvd: 95
```

Live DNS Traffic Capture Validation

The live DNS capture was successfully validated using capinfos and contained six packets exchanged over a 0.451-second interval. The traffic consisted of three DNS queries and three corresponding responses between the host at 192.168.232.128 and the DNS resolver 8.8.8.8. The captured queries were for the A, AAAA, and MX record types of example.com, with each query receiving a corresponding response. The PCAPNG metadata confirms that the traffic was captured on eth0 using a port 53 capture filter, providing a valid packet-level dataset for subsequent DNS forensic analysis.



```
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
└─$ sudo capinfos /tmp/fresh_dig_dns.pcapng
File name: /tmp/fresh_dig_dns.pcapng
File type: Wireshark/... - pcapng
File encapsulation: Ethernet
File timestamp precision: nanoseconds (9)
Packet size limit: file hdr: (not set)
Number of packets: 6
File size: 1,140 bytes
Data size: 595 bytes
Capture duration: 0.451421448 seconds
Earliest packet time: 2026-09-21 15:34:04.993284628
Latest packet time: 2026-09-21 15:34:05.444706076
Data byte rate: 1,318 bytes/s
Data bit rate: 10 kbps
Average packet size: 99.17 bytes
Average packet rate: 13 packets/s
SHA256: a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedcd1d4ee28904
SHA1: 43361beff6aa1b893b43a79bd0b4eaa068339163
Strict time order: True
Capture hardware: Intel(R) Core(TM) i7-4600U CPU @ 2.10GHz (with SSE4.2)
Capture oper-sys: Linux 6.12.33+kali-amd64
Capture application: Dumpcap (Wireshark) 4.6.6
Number of interfaces in file: 1
Interface #0 info:
    Name = eth0
    Encapsulation = Ethernet (1 - ether)
    Capture length = 262144
    Time precision = nanoseconds (9)
    Time ticks per second = 1000000000
    Time resolution = 0x09
    Filter string = port 53
    Operating system = Linux 6.12.33+kali-amd64
    Number of stat entries = 1
    Number of packets = 6

(kali@kali)~[~/SBT-DF203-Lab7]
└─$ sudo tshark -r /tmp/fresh_dig_dns.pcapng -Y dns
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
  1 0.000000000 192.168.232.128 → 8.8.8.8      DNS 82 Standard query 0x253e A example.com OPT
  2 0.145376540      8.8.8.8 → 192.168.232.128 DNS 114 Standard query response 0x253e A example.com A 104.20.23.154 A 172.66.147.243 OPT
  3 0.206090732 192.168.232.128 → 8.8.8.8      DNS 82 Standard query 0x32af AAAA example.com OPT
  4 0.207152419      8.8.8.8 → 192.168.232.128 DNS 138 Standard query response 0x32af AAAA example.com AAAA 2606:4700:10::ac42:93f3 AAAA 2606:4700:10::6814:179a OPT
  5 0.359726924 192.168.232.128 → 8.8.8.8      DNS 82 Standard query 0x5ea3 MX example.com OPT
  6 0.451421448      8.8.8.8 → 192.168.232.128 DNS 97 Standard query response 0x5ea3 MX example.com MX 0 <Root> OPT
```

Live DNS Capture Evidence Preservation

Following validation of the live DNS capture, the PCAPNG file was transferred from the temporary capture location into the designated evidence directory. File ownership was assigned to the kali user to allow controlled analysis, while the original capture contents remained unchanged. A SHA-256 hash was calculated for the preserved evidence file, producing 7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedcd1d4ee28904. The preserved file was confirmed to exist in the evidence directory with a size of approximately 1.2 KB. The recorded hash provides an integrity reference for subsequent forensic examination.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo cp --preserve=timestamps /tmp/fresh_dig_dns.pcapng evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo chown kali:kali evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_capture_hash.txt
a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904 evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ ls -lh evidence/fresh_dig_dns.pcapng
-rw-r--r-- 1 kali kali 1.2K Sep 21 15:34 evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

DNS packet field extraction

The preserved DNS capture was examined using TShark to extract relevant packet-level fields for forensic analysis. The extraction identified three DNS query/response transactions between 192.168.232.128 and 8.8.8.8, with transaction IDs 0x253e, 0x32af, and 0x5ea3. The transactions corresponded to A, AAAA, and MX queries for example.com, respectively. The A response returned the IPv4 addresses 104.20.23.154 and 172.66.147.243, while the AAAA response returned 2606:4700:10::ac42:93f3 and 2606:4700:10::6814:179a, each with a 300-second TTL. The MX transaction was identified by query type 15 and its response was successfully matched to the corresponding query using transaction ID 0x5ea3. The extracted fields provide a structured representation of the DNS activity contained in the preserved capture.

```
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-E frame.number \
-E frame.time \
-E ip.src \
-E ip.dst \
-E dns.id \
-E dns.flags.response \
-E dns.qry.name \
-E dns.qry.type \
-E dns.resp.name \
-E dns.resp.type \
-E dns.resp.ttl \
-E dns.a \
-E dns.aaaa
Warning: program compiled against libxml 215 using older 214
frame.number|frame.time|ip.src|ip.dst|dns.id|dns.flags.response|dns.qry.name|dns.qry.type|dns.resp.name|dns.resp.type|dns.resp.ttl|dns.a|dn
1|2026-09-21T15:34:04.993284628-0400|192.168.232.128|8.8.8.8|0x253e|False|example.com|1|<Root>|41|||
2|2026-09-21T15:34:05.138661168-0400|8.8.8.8|192.168.232.128|0x253e|True|example.com|1|example.com,example.com,<Root>|1,1,41|300,300|104.20
3|2026-09-21T15:34:05.199375360-0400|192.168.232.128|8.8.8.8|0x32af|False|example.com|28|<Root>|41|||
4|2026-09-21T15:34:05.290437047-0400|8.8.8.8|192.168.232.128|0x32af|True|example.com|28|example.com,example.com,<Root>|28,28,41|300,300|26
4:179a
5|2026-09-21T15:34:05.353011552-0400|192.168.232.128|8.8.8.8|0x5ea3|False|example.com|15|<Root>|41|||
6|2026-09-21T15:34:05.444706076-0400|8.8.8.8|192.168.232.128|0x5ea3|True|example.com|15|example.com,<Root>|15,41|300||
```

DNS Packet-Level Summary

The preserved DNS evidence was further examined using TShark to produce a structured packet-level summary of the captured DNS traffic. The extraction records the relative timing, source and destination addresses, DNS transaction IDs, query and response indicators, queried domain, query type, response type, TTL values, and returned IPv4 and IPv6 addresses. The results show three DNS queries for example.com from 192.168.232.128 to 8.8.8.8, followed by corresponding responses. The A transaction returned two IPv4 addresses, while the AAAA transaction returned two IPv6 addresses, with each address record showing a 300-second TTL. The MX transaction is identified by query type 15 and transaction ID 0x5ea3, with its corresponding response captured as packet 6. This extraction provides a concise, repeatable representation of the DNS activity contained in the preserved PCAPNG evidence.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e dns.id \
-e dns.flags.response \
-e dns.qry.name \
-e dns.qry.type \
-e dns.resp.name \
-e dns.resp.type \
-e dns.resp.ttl \
-e dns.a \
-e dns.aaaa
Warning: program compiled against libxml 215 using older 214
frame.number|frame.time|ip.src|ip.dst|dns.id|dns.flags.response|dns.qry.name|dns.qry.type|dns.resp.name|dns.resp.type|dns.resp.ttl|dns.a|dns.aaaa
1|2026-09-21T15:34:04.993284628-0400|192.168.232.128|8.8.8.8|0x253e|False|example.com|1|<Root>|41|||
2|2026-09-21T15:34:05.138661168-0400|8.8.8.8|192.168.232.128|0x253e|True|example.com|1|example.com,example.com,<Root>|1,1,41|300,300|104.20
3|2026-09-21T15:34:05.199375360-0400|192.168.232.128|8.8.8.8|0x32af|False|example.com|28|<Root>|41|||
4|2026-09-21T15:34:05.290437047-0400|8.8.8.8|192.168.232.128|0x32af|True|example.com|28|example.com,example.com,<Root>|28,28,41|300,300||26
4:179a
5|2026-09-21T15:34:05.353011552-0400|192.168.232.128|8.8.8.8|0x5ea3|False|example.com|15|<Root>|41|||
6|2026-09-21T15:34:05.444706076-0400|8.8.8.8|192.168.232.128|0x5ea3|True|example.com|15|example.com,<Root>|15,41|300||
```

DNS Response Status and Answer Counts

The DNS transactions in the preserved capture were examined further to determine response status, recursion availability, and the number of records returned. The three DNS responses have an RCODE of 0, indicating successful DNS responses with no error reported in the response code. Recursion Available was set to True for each response from 8.8.8.8. The A and AAAA responses each contained two answer records, while the MX response contained one answer record. No authoritative answer records were present in the captured responses, while one additional record was reported for each response. The corresponding query packets contained zero answer records, as expected

for DNS requests. The transaction IDs confirm that each response corresponds to its preceding query.

```
(kali@kali) - [~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e frame.time_relative \
-e ip.src \
-e ip.dst \
-e dns.id \
-e dns.flags.response \
-e dns.qry.name \
-e dns.qry.type \
-e dns.resp.type \
-e dns.resp.ttl \
-e dns.a \
-e dns.aaaa \
| tee reports/fresh_dns_packet_summary.txt
Warning: program compiled against libxml 215 using older 214
frame.number|frame.time_relative|ip.src|ip.dst|dns.id|dns.flags.response|dns.qry.name|dns.qry.type|dns.resp.type|dns.resp.ttl|dns.a|dns.aaaa
1|0.000000000|192.168.232.128|8.8.8.8|0x253e|False|example.com|1|41|1|
2|0.145376540|8.8.8.8|192.168.232.128|0x253e|True|example.com|1|1,41|300,300|104.20.23.154,172.66.147.243|
3|0.206090732|192.168.232.128|8.8.8.8|0x32af|False|example.com|28|41|1|
4|0.297152419|8.8.8.8|192.168.232.128|0x32af|True|example.com|28|28,41|300,300|2606:4700:10::ac42:93f3,2606:4700:10::6814:179a|
5|0.359726924|192.168.232.128|8.8.8.8|0x5ea3|False|example.com|15|41|1|
6|0.451421448|8.8.8.8|192.168.232.128|0x5ea3|True|example.com|15|15,41|300||
```

DNS Flags and Recursion Analysis

The DNS headers and control flags were examined to establish the communication characteristics of the captured transactions. All three client queries used standard DNS opcode 0 and had the Recursion Desired (RD) flag set, indicating that the client requested recursive resolution. The corresponding responses from 8.8.8.8 had the Recursion Available (RA) flag set, confirming that the resolver indicated support for recursive queries. Each response had an RCODE of 0, indicating a successful response. The transaction pairs maintained matching DNS identifiers, with 0x253e for the A query, 0x32af for the AAAA query, and 0x5ea3 for the MX query. The response counts further show two answers for the A and AAAA requests and one answer for the MX request, with no authoritative records returned in the captured responses.

```
File Actions Edit View Help
5|0x5ea3|False|1|1|example.com|15|0|0|1
6|0x5ea3|True|0|True|1|1|example.com|15|1|0|1

(kali@kali) - [~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e dns.id \
-e dns.flags \
-e dns.flags.response \
-e dns.flags.opcode \
-e dns.flags.recdesired \
-e dns.flags.recavail \
-e dns.flags.ad \
-e dns.flags.rcode \
-e dns.qry.name \
-e dns.qry.type \
-e dns.count.queries \
-e dns.count.answers \
-e dns.count.auth_rr \
-e dns.count.add_rr
Warning: program compiled against libxml 215 using older 214
frame.number|dns.id|dns.flags|dns.flags.response|dns.flags.opcode|dns.flags.recdesired|dns.flags.recavail|dns.flags.ad|dns.flags.rcode|dns.qry.name|dns.qry.type|dns.count.queries|dns.count.answers|dns.count.auth_rr|dns.count.add_rr
1|0x253e|0x0100|False|0|True|1|1|example.com|1|1|0|0|1
2|0x253e|0x8180|True|0|True|True|0|0|example.com|1|1|2|0|1
3|0x32af|0x0100|False|0|True|1|1|example.com|28|1|0|0|1
4|0x32af|0x8180|True|0|True|True|0|0|example.com|28|1|2|0|1
5|0x5ea3|0x0100|False|0|True|1|1|example.com|15|1|0|0|1
6|0x5ea3|0x8180|True|0|True|True|0|0|example.com|15|1|1|0|1
```

EDNS Field Extraction Validation

An attempt was made to extract EDNS-related fields from the preserved DNS capture using TShark. The extraction did not execute because several specified EDNS field names were not recognized by the installed TShark/Wireshark version, including `dns.edns.version`, `dns.edns.udp_payload_size`, `dns.edns.ext_rcode`, and `dns.edns.z`. This result indicates that the selected field identifiers are not available in the current protocol-field definition and does not indicate an issue with the underlying PCAPNG evidence. No forensic conclusion was drawn from this unsuccessful extraction attempt, and the preserved capture remained unchanged.

```
(kali㉿kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
-e dns.resp.type \
-e dns.flags.response \
-e dns.edns.version \
-e dns.edns.udp_payload_size \
-e dns.edns.ext_rcode \
-e dns.edns.z \
-e dns.resp.z.do
Warning: program compiled against libxml 215 using older 214
tshark: Some fields aren't valid:
  dns.edns.z
  dns.edns.ext_rcode
  dns.edns.udp_payload_size
  dns.edns.version
```

EDNS and DNSSEC Field Analysis

The EDNS-related fields were successfully extracted from the DNS capture using field names supported by the installed TShark version. The analysis showed EDNS version 0 across all six DNS packets. The advertised UDP payload size was 1472 bytes in the query packets and 512 bytes in the corresponding responses. The DNSSEC OK (DO) bit was not set in any of the captured packets. No specific EDNS option code, length, or option data was exposed in the extracted fields. These observations provide additional protocol-level context for the captured DNS transactions without, by themselves, indicating anomalous DNS activity.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
-e dns.flags.response \
-e dns.resp.edns0_version \
-e dns.rr.udp_payload_size \
-e dns.resp.z.do \
-e dns.opt.code \
-e dns.opt.len \
-e dns.opt.data
Warning: program compiled against libxml 215 using older 214
frame.number|dns.id|dns.qry.name|dns.qry.type|dns.flags.response|dns.resp.edns0_version|dns.rr.udp_payload_size|dns.resp.z.do|dns.opt.code|
dns.opt.len|dns.opt.data
1|0x253e|example.com|1|False|0|1472|False|||
2|0x253e|example.com|1|True|0|512|False|||
3|0x32af|example.com|28|False|0|1472|False|||
4|0x32af|example.com|28|True|0|512|False|||
5|0x5ea3|example.com|15|False|0|1472|False|||
6|0x5ea3|example.com|15|True|0|512|False|||

```

Detailed DNS Query and EDNS OPT Analysis

A detailed protocol-level inspection was performed on the first captured DNS packet to validate the EDNS information identified during field extraction. The packet was an A-record query for example.com sent from the local host (192.168.232.128) to the upstream DNS server (8.8.8.8) over UDP port 53, with transaction ID 0x253e. The query requested recursive resolution and contained one question with no answer or authority records. One additional record was present in the form of an EDNS OPT pseudo-record. The OPT record specified EDNS version 0, a UDP payload size of 1472 bytes, an extended RCODE value of 0, and a DNSSEC OK (DO) bit that was not set. The EDNS data length was 0, indicating that no additional EDNS option data was carried in this query.

```

nub4 - VMware Workstation
File Actions Edit View Help
6|0x5ea3|example.com|15|True|0|512|False|||

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'frame.number == 1' \
-V
Warning: program compiled against libxml 215 using older 214
Frame 1: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface eth0, id 0
  Section number: 1
    Interface id: 0 (eth0)
      Interface name: eth0
        Encapsulation type: Ethernet (1)
          Arrival Time: Sep 21, 2026 15:34:04.993284628 EDT
            UTC Arrival Time: Sep 21, 2026 19:34:04.993284628 UTC
              Epoch Arrival Time: 1790019244.993284628
                [Time shift for this packet: 0.000000000 seconds]
                  [Time since reference or first frame: 0.000000000 seconds]
                    Frame Number: 1
                      Frame Length: 82 bytes (656 bits)
                        Capture Length: 82 bytes (656 bits)
                          [Frame is marked: False]
                            [Frame is ignored: False]
                              [Protocols in frame: eth:ethertype:ip:udp:dns]
                                Character encoding: ASCII (0)
                                  Ethernet II, Src: VMware_77:75:3c (00:0c:29:77:75:3c), Dst: VMware_fe:76:b0 (00:50:56:fe:76:b0)
                                    Destination: VMware_fe:76:b0 (00:50:56:fe:76:b0)
                                      .... 0. .... = LG bit: Globally unique address (factory default)
                                      .... 0 .... = IG bit: Individual address (unicast)
                                    Source: VMware_77:75:3c (00:0c:29:77:75:3c)
                                      .... 0. .... = LG bit: Globally unique address (factory default)
                                      .... 0 .... = IG bit: Individual address (unicast)
                                    Type: IPv4 (0x0800)
                                      [Stream index: 0]
                                    Internet Protocol Version 4, Src: 192.168.232.128, Dst: 8.8.8.8
                                      0100 .... = Version: 4
                                      .... 0101 = Header Length: 20 bytes (5)

```

```
md64 - VMware Workstation
Help
kali-linux-2025.2-vmware... X
1 2 3 4
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 68
Identification: 0x7885 (30853)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
..0. .... = Don't fragment: Not set
...0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0x48eb [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.232.128
Destination Address: 8.8.8.8
[Stream index: 0]
User Datagram Protocol, Src Port: 42371, Dst Port: 53
Source Port: 42371
Destination Port: 53
Length: 48
Checksum: 0xb97a [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 1]
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (40 bytes)
Domain Name System (query)
Transaction ID: 0x253e
Flags: 0x0100 Standard query
0... .... = Response: Message is a query
.000 0... .... = Opcode: Standard query (0)
.... ..0. .... = Truncated: Message is not truncated
.... ...1. .... = Recursion desired: Do query recursively
.... .... .0.. .... = Z: reserved (0)
```

```
md64 - VMware Workstation
Help
kali-linux-2025.2-vmware... X
1 2 3 4
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (40 bytes)
Domain Name System (query)
Transaction ID: 0x253e
Flags: 0x0100 Standard query
0... .... = Response: Message is a query
.000 0... .... = Opcode: Standard query (0)
.... ..0. .... = Truncated: Message is not truncated
.... ...1. .... = Recursion desired: Do query recursively
.... .... .0.. .... = Z: reserved (0)
.... .... ...0 .... = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
example.com: type A, class IN
Name: example.com
[Name Length: 11]
[Label Count: 2]
Type: A (1) (Host Address)
Class: IN (0x0001)
Additional records
<Root>: type OPT
Name: <Root>
Type: OPT (41)
UDP payload size: 1472
Higher bits in extended RCODE: 0x00
EDNS0 version: 0
Z: 0x0000
0... .... = DO bit: Cannot handle DNSSEC security RRs
.000 0000 0000 0000 = Reserved: 0x0000
Data length: 0
```

DNS Query and EDNS Details

Field	Observed Value	Forensic Significance
Source IP	192.168.232.128	Local host that generated the DNS query
Destination IP	8.8.8.8	Upstream DNS resolver receiving the query
UDP Destination Port	53	Standard DNS service port
DNS Transaction ID	0x253e	Identifies and correlates the query with its response
Query Name	example.com	Domain requested by the host
Query Type	A	Requests an IPv4 address for the domain
Recursion Desired (RD)	Set	The client requested recursive resolution
Answer Records	0	Expected for an outbound DNS query
Authority Records	0	No authority records included in the query
Additional Records	1	One additional record accompanies the query
Additional Record Type	OPT (41)	EDNS OPT pseudo-record
EDNS Version	0	EDNS version used by the query
UDP Payload Size	1472 bytes	Maximum UDP DNS payload size advertised by the client
Extended RCODE	0	No extended DNS error code indicated
DNSSEC OK (DO) Bit	Not set	The query did not request DNSSEC records
EDNS Option Data Length	0	No additional EDNS option data was carried


```
kali-linux-2025.2-vmware... x
File Actions Edit View Help
..... 0000 = Reply code: No error (0)
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 1
Queries
  example.com: type A, class IN
    Name: example.com
    [Name Length: 11]
    [Label Count: 2]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
Answers
  example.com: type A, class IN, addr 104.20.23.154
    Name: example.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 300 (5 minutes)
    Data length: 4
    Address: 104.20.23.154
  example.com: type A, class IN, addr 172.66.147.243
    Name: example.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 300 (5 minutes)
    Data length: 4
    Address: 172.66.147.243
Additional records
  <Root>: type OPT
    Name: <Root>
    Type: OPT (41)
    UDP payload size: 512
    Higher bits in extended RCODE: 0x00
    EDNS0 version: 0
    Z: 0x0000
    0 ... .. = DO bit: Cannot handle DNSSEC security RRs
```

Detailed DNS Response and EDNS OPT Analysis

Packet 2 was examined as the response corresponding to the A-record query in Packet 1. The DNS transaction ID of 0x253e matched the preceding query, confirming the request-response relationship. The response originated from 8.8.8.8 and was delivered to the local host at 192.168.232.128 through UDP port 53. The resolver returned a successful response with RCODE 0 and provided two A records for example.com: 104.20.23.154 and 172.66.147.243, each with a TTL of 300 seconds. The response was not authoritative and was not truncated, while recursion was both requested and available. One EDNS OPT pseudo-record was present, specifying EDNS version 0, a UDP payload size of 512 bytes, an extended RCODE of 0, and an unset DNSSEC OK (DO) bit. No additional EDNS option data was present. The measured response interval from the captured query was approximately 145.38 milliseconds.


```
64 - VMware Workstation
Help
kali-linux-2025.2-vmware...
kali@kali: ~/SBT-DF203-Lab7

File Actions Edit View Help

... .. 0. .... = Answer authenticated: Answer/authority portion was not authenticated by the server
... .. 0. .... = Non-authenticated data: Unacceptable
... .. 0000 = Reply code: No error (0)

Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 1
Queries
  example.com: type A, class IN
    Name: example.com
    [Name Length: 11]
    [Label Count: 2]
    Type: A (1) (Host Address)
    Class: IN (0x0001)

Answers
  example.com: type A, class IN, addr 104.20.23.154
    Name: example.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 300 (5 minutes)
    Data length: 4
    Address: 104.20.23.154
  example.com: type A, class IN, addr 172.66.147.243
    Name: example.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 300 (5 minutes)
    Data length: 4
    Address: 172.66.147.243

Additional records
  <Root>: type OPT
    Name: <Root>
    Type: OPT (41)
    UDP payload size: 512
    Higher bits in extended RCODE: 0x00
    EDNS0 version: 0
    Z: 0x0000
    0... .. = DO bit: Cannot handle DNSSEC security RRs
    .000 0000 0000 0000 = Reserved: 0x0000
    Data length: 0

[Request In: 1]
[Time: 145.376540 milliseconds]
```

DNS Response and EDNS Details

Field	Observed Value	Forensic Significance
Source IP	8.8.8.8	DNS resolver responding to the query
Destination IP	192.168.232.128	Local host receiving the response
UDP Source Port	53	Standard DNS service port
UDP Destination Port	42371	Ephemeral port used by the requesting host
DNS Transaction ID	0x253e	Matches Packet 1 and links this response to the query
Response Status	No error (RCODE 0)	Resolver successfully processed the request
Authoritative	Not authoritative	The responding server is not authoritative for example.com

Field	Observed Value	Forensic Significance
Truncated	No	Response was not truncated
Recursion Desired (RD)	Set	The original client requested recursive resolution
Recursion Available (RA)	Set	The resolver supports recursive queries
Answer Authenticated	Not authenticated	DNSSEC authentication was not indicated
Query Name	example.com	Matches the original query
Query Type	A	IPv4 address lookup
Answer Records	2	Two IPv4 addresses were returned
Answer 1	104.20.23.154	IPv4 address returned for example.com
Answer 2	172.66.147.243	IPv4 address returned for example.com
TTL	300 seconds	Both A records were valid for 5 minutes at the time of capture
Authority Records	0	No authority section records were returned
Additional Records	1	One EDNS OPT pseudo-record was included
EDNS Version	0	EDNS version used in the response
UDP Payload Size	512 bytes	UDP payload size indicated by the responding resolver
Extended RCODE	0	No extended DNS error code
DNSSEC OK (DO) Bit	Not set	DNSSEC records were not requested/indicated
EDNS Option Data Length	0	No additional EDNS option data was carried
Response Time	145.376540 ms	Time between the captured query and corresponding response

Detailed AAAA DNS Query and EDNS OPT Analysis

Packet 3 was examined as the second DNS query in the captured exchange. The packet originated from 192.168.232.128 and was sent to the upstream DNS resolver at 8.8.8.8 over UDP port 53, using transaction ID 0x32af. The query requested an AAAA record for example.com, indicating a request for an IPv6 address, and the Recursion Desired flag was set. No answer or authority records were present in the query, while one additional OPT pseudo-record was included for EDNS. The OPT record specified EDNS version 0, a UDP payload size of 1472 bytes, an extended RCODE value of 0, and an unset DNSSEC OK (DO) bit. The EDNS data length was 0, indicating that no additional EDNS option data was carried in the query.

```

kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
L$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'frame.number == 3' \
-v
Warning: program compiled against libxml 215 using older 214
Frame 3: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface eth0, id 0
  Section number: 1
  Interface id: 0 (eth0)
    Interface name: eth0
  Encapsulation type: Ethernet (1)
  Arrival Time: Sep 21, 2026 15:34:05.199375360 EDT
  UTC Arrival Time: Sep 21, 2026 19:34:05.199375360 UTC
  Epoch Arrival Time: 1790019245.199375360
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 60.714192 milliseconds]
  [Time since reference or first frame: 206.090732 milliseconds]
  Frame Number: 3
  Frame Length: 82 bytes (656 bits)
  Capture Length: 82 bytes (656 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:udp:dns]
  Character encoding: ASCII (0)
  Ethernet II, Src: VMware_77:75:3c (00:0c:29:77:75:3c), Dst: VMWare_fe:76:b0 (00:50:56:fe:76:b0)
    Destination: VMWare_fe:76:b0 (00:50:56:fe:76:b0)
      .... 0. .... = LG bit: Globally unique address (factory default)
      .... 0 .... = IG bit: Individual address (unicast)
    Source: VMware_77:75:3c (00:0c:29:77:75:3c)
      .... 0. .... = LG bit: Globally unique address (factory default)
      .... 0 .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 0]
  Internet Protocol Version 4, Src: 192.168.232.128, Dst: 8.8.8.8
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 68
    Identification: 0x8aaa (35498)
    000. .... = Flags: 0x0
      0... .... = Reserved bit: Not set
      .0.. .... = Don't fragment: Not set
      ..0. .... = More fragments: Not set

```

```

d64 - VMware Workstation
Help
kali-linux-2025.2-vmware...
kali@kali: ~/SBT-DF203-Lab7

File Actions Edit View Help
Protocol: UDP (17)
Header Checksum: 0x36c6 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.232.128
Destination Address: 8.8.8.8
[Stream index: 0]
User Datagram Protocol, Src Port: 34860, Dst Port: 53
Source Port: 34860
Destination Port: 53
Length: 48
Checksum: 0xb97a [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (40 bytes)
Domain Name System (query)
Transaction ID: 0x32af
Flags: 0x0100 Standard query
0 ... .. = Response: Message is a query
.000 0 ... .. = Opcode: Standard query (0)
... ..0. ... .. = Truncated: Message is not truncated
... ..1 ... .. = Recursion desired: Do query recursively
... ..0.. ... .. = Z: reserved (0)
... ..0 ... .. = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
example.com: type AAAA, class IN
Name: example.com
[Name Length: 11]
[Label Count: 2]
Type: AAAA (28) (IP6 Address)
Class: IN (0x0001)
Additional records
<Root>: type OPT
Name: <Root>
Type: OPT (41)
UDP payload size: 1472

```

```

d64 - VMware Workstation
Help
kali-linux-2025.2-vmware...
kali@kali: ~/SBT-DF203-Lab7

File Actions Edit View Help
Checksum: 0xb97a [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (40 bytes)
Domain Name System (query)
Transaction ID: 0x32af
Flags: 0x0100 Standard query
0 ... .. = Response: Message is a query
.000 0 ... .. = Opcode: Standard query (0)
... ..0. ... .. = Truncated: Message is not truncated
... ..1 ... .. = Recursion desired: Do query recursively
... ..0.. ... .. = Z: reserved (0)
... ..0 ... .. = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
example.com: type AAAA, class IN
Name: example.com
[Name Length: 11]
[Label Count: 2]
Type: AAAA (28) (IP6 Address)
Class: IN (0x0001)
Additional records
<Root>: type OPT
Name: <Root>
Type: OPT (41)
UDP payload size: 1472
Higher bits in extended RCODE: 0x00
EDNS0 version: 0
Z: 0x0000
0 ... .. = DO bit: Cannot handle DNSSEC security RRs
.000 0000 0000 0000 = Reserved: 0x0000
Data length: 0

(kali@kali)~[~/SBT-DF203-Lab7]
$

```

DNS AAAA Query and EDNS Details

Field	Observed Value	Forensic Significance
Source IP	192.168.232.128	Local host that generated the DNS query
Destination IP	8.8.8.8	Upstream DNS resolver receiving the query
UDP Source Port	34860	Ephemeral port used by the local host
UDP Destination Port	53	Standard DNS service port
DNS Transaction ID	0x32af	Identifies this DNS transaction and links it to its response
Query Name	example.com	Domain requested by the host
Query Type	AAAA	Requests an IPv6 address for the domain
Recursion Desired (RD)	Set	The client requested recursive resolution
Answer Records	0	Expected for an outbound DNS query
Authority Records	0	No authority records included
Additional Records	1	One additional record accompanies the query

Field	Observed Value	Forensic Significance
Additional Record Type	OPT (41)	EDNS OPT pseudo-record
EDNS Version	0	EDNS version used by the query
UDP Payload Size	1472 bytes	UDP payload size advertised by the client
Extended RCODE	0	No extended DNS error code indicated
DNSSEC OK (DO) Bit	Not set	DNSSEC records were not requested
EDNS Option Data Length	0	No additional EDNS option data was carried
Frame Time	206.090732 ms from first frame	Position of the packet within the captured DNS exchange

Detailed AAAA DNS Response and EDNS OPT Analysis

Packet 4 was examined as the response corresponding to the AAAA query in Packet 3. The transaction ID of 0x32af matched the preceding query, confirming the request-response relationship, while the source and destination addresses and UDP ports were appropriately reversed. The resolver returned a successful response with RCODE 0 and provided two IPv6 addresses for example.com: 2606:4700:10::ac42:93f3 and 2606:4700:10::6814:179a, each with a TTL of 300 seconds. The response was not authoritative or truncated, and both recursion desired and recursion available were indicated. An EDNS OPT pseudo-record was also present, specifying EDNS version 0, a UDP payload size of 512 bytes, an extended RCODE of 0, and an unset DNSSEC OK (DO) bit. The EDNS data length was 0, indicating that no additional EDNS option data

```

kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'frame.number == 4' \
-v
Warning: program compiled against libxml 215 using older 214
Frame 4: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface eth0, id 0
  Section number: 1
    Interface id: 0 (eth0)
      Interface name: eth0
        Encapsulation type: Ethernet (1)
        Arrival Time: Sep 21, 2026 15:34:05.290437047 EDT
        UTC Arrival Time: Sep 21, 2026 19:34:05.290437047 UTC
        Epoch Arrival Time: 1790019245.290437047
        [Time shift for this packet: 0.000000000 seconds]
        [Time delta from previous captured frame: 91.061687 milliseconds]
        [Time since reference or first frame: 297.152419 milliseconds]
        Frame Number: 4
        Frame Length: 138 bytes (1104 bits)
        Capture Length: 138 bytes (1104 bits)
        [Frame is marked: False]
        [Frame is ignored: False]
        [Protocols in frame: eth:ethertype:ip:udp:dns]
        Character encoding: ASCII (0)
      Ethernet II, Src: VMware_fe:76:b0 (00:50:56:fe:76:b0), Dst: VMware_77:75:3c (00:0c:29:77:75:3c)
        Destination: VMware_77:75:3c (00:0c:29:77:75:3c)
          .... ..0. .... = LG bit: Globally unique address (factory default)
          .... ..0. .... = IG bit: Individual address (unicast)
        Source: VMware_fe:76:b0 (00:50:56:fe:76:b0)
          .... ..0. .... = LG bit: Globally unique address (factory default)
          .... ..0. .... = IG bit: Individual address (unicast)
        Type: IPv4 (0x0800)
        [Stream index: 0]
      Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.232.128
        0100 .... = Version: 4
        .... 0101 = Header Length: 20 bytes (5)
        Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
          0000 00.. = Differentiated Services Codepoint: Default (0)
          .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
        Total Length: 124
        Identification: 0x5d91 (23953)
        000. .... = Flags: 0x0
          0... .... = Reserved bit: Not set

```

```
664 - VMware Workstation
Help | [Icons] | [Taskbar] | [Terminal] | [VM] | [Host] | [Menu]

kali-linux-2025.2-vmware...X
[Icons] | [Taskbar] | [Terminal] | [VM] | [Host] | [Menu]

kali@kali: ~/SBT-DF203-Lab7

File Actions Edit View Help

0 ... .. = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0x23a7 [validation disabled]
[Header checksum status: Unverified]
Source Address: 8.8.8.8
Destination Address: 192.168.232.128
[Stream index: 0]
User Datagram Protocol, Src Port: 53, Dst Port: 34860
Source Port: 53
Destination Port: 34860
Length: 104
Checksum: 0x409d [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 2]
[Timestamps]
[Time since first frame: 91.061687 milliseconds]
[Time since previous frame: 91.061687 milliseconds]
UDP payload (96 bytes)
Domain Name System (response)
Transaction ID: 0x32af
Flags: 0x8180 Standard query response, No error
1 ... .. = Response: Message is a response
.000 0 ... .. = Opcode: Standard query (0)
.... 0. .... = Authoritative: Server is not an authority for domain
.... 0. .... = Truncated: Message is not truncated
.... 1 ... .. = Recursion desired: Do query recursively
.... 1 ... .. = Recursion available: Server can do recursive queries
.... 0 ... .. = Z: reserved (0)
.... 0. .... = Answer authenticated: Answer/authority portion was not authenticated by the server
.... 0 ... .. = Non-authenticated data: Unacceptable
.... 0 ... .. = Reply code: No error (0)
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 1
Queries
example.com: type AAAA, class IN
Name: example.com
```

```
md64 - VMware Workstation
Help | [Icons] | [Taskbar] | [Terminal] | [VM] | [Host] | [Menu]

kali-linux-2025.2-vmware...X
[Icons] | [Taskbar] | [Terminal] | [VM] | [Host] | [Menu]

kali@kali: ~/SBT-DF203-Lab7

File Actions Edit View Help

Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 1
Queries
example.com: type AAAA, class IN
Name: example.com
[Name Length: 11]
[Label Count: 2]
Type: AAAA (28) (IPv6 Address)
Class: IN (0x0001)
Answers
example.com: type AAAA, class IN, addr 2606:4700:10::ac42:93f3
Name: example.com
Type: AAAA (28) (IPv6 Address)
Class: IN (0x0001)
Time to live: 300 (5 minutes)
Data length: 16
AAAA Address: 2606:4700:10::ac42:93f3
example.com: type AAAA, class IN, addr 2606:4700:10::6814:179a
Name: example.com
Type: AAAA (28) (IPv6 Address)
Class: IN (0x0001)
Time to live: 300 (5 minutes)
Data length: 16
AAAA Address: 2606:4700:10::6814:179a
Additional records
<Root>: type OPT
Name: <Root>
Type: OPT (41)
UDP payload size: 512
Higher bits in extended RCODE: 0x00
EDNS0 version: 0
Z: 0x0000
0 ... .. = DO bit: Cannot handle DNSSEC security RRs
.000 0000 0000 0000 = Reserved: 0x0000
Data length: 0
[Request In: 3]
[Time: 91.061687 milliseconds]

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

DNS AAAA Response and EDNS Details

Field	Observed Value	Forensic Significance
Source IP	8.8.8.8	DNS resolver responding to the query
Destination IP	192.168.232.128	Local host receiving the response
UDP Source Port	53	Standard DNS service port
UDP Destination Port	34860	Matches the source port used in Packet 3
DNS Transaction ID	0x32af	Matches Packet 3 and confirms the request-response relationship
Response Status	No error (RCODE 0)	Query was successfully processed
Authoritative	No	Resolver is not authoritative for example.com
Truncated	No	Complete response was received
Recursion Desired (RD)	Set	Recursive resolution was requested
Recursion Available (RA)	Set	Resolver supports recursive resolution
Answer Authenticated	Not authenticated	DNSSEC authentication was not indicated
Query Name	example.com	Matches Packet 3

Field	Observed Value	Forensic Significance
Query Type	AAAA	IPv6 address lookup
Answer Records	2	Two IPv6 addresses were returned
Answer 1	2606:4700:10::ac42:93f3	IPv6 address returned
Answer 2	2606:4700:10::6814:179a	IPv6 address returned
TTL	300 seconds	Both AAAA records have a 5-minute TTL
Authority Records	0	No authority records were returned
Additional Records	1	One EDNS OPT pseudo-record was included
EDNS Version	0	EDNS version used in the response
UDP Payload Size	512 bytes	UDP payload size indicated by the resolver
Extended RCODE	0	No extended DNS error code
DNSSEC OK (DO) Bit	Not set	DNSSEC records were not requested
EDNS Option Data Length	0	No additional EDNS option data was carried
Response Time	91.061687 ms	Time from Packet 3 to Packet 4

Detailed MX DNS Query and EDNS OPT Analysis

Packet 5 was examined as the third DNS query in the captured exchange. The packet originated from 192.168.232.128 and was sent to the upstream DNS resolver at 8.8.8.8 over UDP port 53, using transaction ID 0x5ea3. The query requested an MX record for example.com and had the Recursion Desired flag set. No answer or authority records were present in the query, while one additional OPT pseudo-record was included for EDNS. The OPT record specified EDNS version 0, a UDP payload size of 1472 bytes, an extended RCODE value of 0, and an unset DNSSEC OK (DO) bit. The EDNS data length was 0, indicating that no additional EDNS option data was carried in the query.

```
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
(kali@kali)~[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'frame.number == 5' \
-V
Warning: program compiled against libxml 215 using older 214
Frame 5: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface eth0, id 0
  Section number: 1
    Interface id: 0 (eth0)
      Interface name: eth0
      Encapsulation type: Ethernet (1)
      Arrival Time: Sep 21, 2026 15:34:05.353011552 EDT
      UTC Arrival Time: Sep 21, 2026 19:34:05.353011552 UTC
      Epoch Arrival Time: 1790019245.353011552
      [Time shift for this packet: 0.000000000 seconds]
      [Time delta from previous captured frame: 62.574505 milliseconds]
      [Time since reference or first frame: 359.726924 milliseconds]
      Frame Number: 5
      Frame Length: 82 bytes (656 bits)
      Capture Length: 82 bytes (656 bits)
      [Frame is marked: False]
      [Frame is ignored: False]
      [Protocols in frame: eth:ethertype:ip:udp:dns]
      Character encoding: ASCII (0)
      Ethernet II, Src: VMware_77:75:3c (00:0c:29:77:75:3c), Dst: VMware_fe:76:b0 (00:50:56:fe:76:b0)
        Destination: VMware_fe:76:b0 (00:50:56:fe:76:b0)
          .... 0. .... = LG bit: Globally unique address (factory default)
          .... 0. .... = IG bit: Individual address (unicast)
        Source: VMware_77:75:3c (00:0c:29:77:75:3c)
          .... 0. .... = LG bit: Globally unique address (factory default)
          .... 0. .... = IG bit: Individual address (unicast)
        Type: IPv4 (0x0800)
        [Stream index: 0]
      Internet Protocol Version 4, Src: 192.168.232.128, Dst: 8.8.8.8
        0100 .... = Version: 4
        .... 0101 = Header Length: 20 bytes (5)
        Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
          0000 00.. = Differentiated Services Codepoint: Default (0)
          .... 00.. = Explicit Congestion Notification: Not ECN-Capable Transport (0)
        Total Length: 68
        Identification: 0x0d26 (3366)
        000. .... = Flags: 0x0
          0... .... = Reserved bit: Not set
          .0.. .... = Don't fragment: Not set
```


Field	Observed Value	Forensic Significance
UDP Destination Port	53	Standard DNS service port
DNS Transaction ID	0x5ea3	Identifies this DNS transaction and links it to its response
Query Name	example.com	Domain requested by the host
Query Type	MX	Requests mail-exchange information for the domain
Recursion Desired (RD)	Set	The client requested recursive resolution
Answer Records	0	Expected for an outbound DNS query
Authority Records	0	No authority records included
Additional Records	1	One additional record accompanies the query
Additional Record Type	OPT (41)	EDNS OPT pseudo-record
EDNS Version	0	EDNS version used by the query
UDP Payload Size	1472 bytes	UDP payload size advertised by the client
Extended RCODE	0	No extended DNS error code indicated

Field	Observed Value	Forensic Significance
DNSSEC OK (DO) Bit	Not set	DNSSEC records were not requested
EDNS Option Data Length	0	No additional EDNS option data was carried
Frame Time	359.726924 ms from first frame	Position of the packet within the captured DNS exchange

Detailed MX DNS Response and EDNS OPT Analysis

Packet 6 was examined as the response corresponding to the MX query in Packet 5. The transaction ID of 0x5ea3 matched the preceding query, confirming the request-response relationship, while the source and destination addresses and UDP ports were reversed as expected. The DNS resolver returned a successful response with RCODE 0 and one MX record for example.com. The returned record had a preference value of 0 and a mail-exchange target represented as the root label (<Root>), with a TTL of 300 seconds. The response was not authoritative or truncated, and recursion was both requested and available. An EDNS OPT pseudo-record was present, specifying EDNS version 0, a UDP payload size of 512 bytes, an extended RCODE of 0, and an unset DNSSEC OK (DO) bit. The EDNS data length was 0, indicating that no additional EDNS option data was carried. The response was received approximately 91.69 milliseconds after the corresponding query.

```
md64 - VMware Workstation
Help
File Actions Edit View Help
kali@kali: ~/SBT-DF203-Lab7
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'frame.number == 6' \
-V
Warning: program compiled against libxml 215 using older 214
Frame 6: Packet, 97 bytes on wire (776 bits), 97 bytes captured (776 bits) on interface eth0, id 0
  Section number: 1
    Interface id: 0 (eth0)
      Interface name: eth0
        Encapsulation type: Ethernet (1)
        Arrival Time: Sep 21, 2026 15:34:05.444706076 EDT
        UTC Arrival Time: Sep 21, 2026 19:34:05.444706076 UTC
        Epoch Arrival Time: 1790019245.444706076
        [Time shift for this packet: 0.000000000 seconds]
        [Time delta from previous captured frame: 91.694524 milliseconds]
        [Time since reference or first frame: 451.421448 milliseconds]
        Frame Number: 6
        Frame Length: 97 bytes (776 bits)
        Capture Length: 97 bytes (776 bits)
        [Frame is marked: False]
        [Frame is ignored: False]
        [Protocols in frame: eth:ethertype:ip:udp:dns]
        Character encoding: ASCII (0)
        Ethernet II, Src: VMware_fe76:b0 (00:50:56:fe:76:b0), Dst: VMware_77:75:3c (00:0c:29:77:75:3c)
          Destination: VMware_77:75:3c (00:0c:29:77:75:3c)
            .... 0... = LG bit: Globally unique address (factory default)
            .... 0... = IG bit: Individual address (unicast)
          Source: VMware_fe76:b0 (00:50:56:fe:76:b0)
            .... 0... = LG bit: Globally unique address (factory default)
            .... 0... = IG bit: Individual address (unicast)
          Type: IPv4 (0x0800)
          [Stream index: 0]
          Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.232.128
            0100 .... = Version: 4
            .... 0101 = Header Length: 20 bytes (5)
            Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
              0000 00.. = Differentiated Services Codepoint: Default (0)
              .... 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
            Total Length: 83
            Identification: 0x5d92 (23954)
            000. .... = Flags: 0x0
              0... .... = Reserved bit: Not set
              .0.. .... = Don't fragment: Not set
```

```
md64 - VMware Workstation
Help
File Actions Edit View Help
kali@kali: ~/SBT-DF203-Lab7
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0x23cf [validation disabled]
[Header checksum status: Unverified]
Source Address: 8.8.8.8
Destination Address: 192.168.232.128
[Stream index: 0]
User Datagram Protocol, Src Port: 53, Dst Port: 40274
Source Port: 53
Destination Port: 40274
Length: 63
Checksum: 0x7758 [unverified]
[Checksum Status: Unverified]
[Stream index: 2]
[Stream Packet Number: 2]
[Timestamps]
  [Time since first frame: 91.694524 milliseconds]
  [Time since previous frame: 91.694524 milliseconds]
  UDP payload (55 bytes)
  Domain Name System (response)
    Transaction ID: 0x5ea3
    Flags: 0x8180 Standard query response, No error
      1... .... = Response: Message is a response
      .000 0... = Opcode: Standard query (0)
      .... 0... = Authoritative: Server is not an authority for domain
      .... 0... = Truncated: Message is not truncated
      .... 1... = Recursion desired: Do query recursively
      .... 1... = Recursion available: Server can do recursive queries
      .... 0... = Z: reserved (0)
      .... 0... = Answer authenticated: Answer/authority portion was not authenticated by the server
      .... 0... = Non-authenticated data: Unacceptable
      .... 0000 = Reply code: No error (0)
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 1
    Queries
      example.com: type MX, class IN
        Name: example.com
        [Name Length: 11]
        [Label Count: 2]
        Type: MX (15) (Mail eXchange)
        Class: IN (0x0001)
```

```
664 - VMware Workstation
Help
kali-linux-2025.2-vmware...
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
... 1 ... = Recursion desired: Do query recursively
... 1... .. = Recursion available: Server can do recursive queries
... .. = Z: reserved (0)
... .. = Answer authenticated: Answer/authority portion was not authenticated by the server
... .. = Non-authenticated data: Unacceptable
... .. 0000 = Reply code: No error (0)
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 1
Queries
  example.com: type MX, class IN
    Name: example.com
    [Name Length: 11]
    [Label Count: 2]
    Type: MX (15) (Mail eXchange)
    Class: IN (0x0001)
Answers
  example.com: type MX, class IN, preference 0, mx <Root>
    Name: example.com
    Type: MX (15) (Mail eXchange)
    Class: IN (0x0001)
    Time to live: 300 (5 minutes)
    Data length: 3
    Preference: 0
    Mail Exchange: <Root>
Additional records
  <Root>: type OPT
    Name: <Root>
    Type: OPT (41)
    UDP payload size: 512
    Higher bits in extended RCODE: 0x00
    EDNS0 version: 0
    Z: 0x0000
    0 ... .. = DO bit: Cannot handle DNSSEC security RRs
    .000 0000 0000 0000 = Reserved: 0x0000
    Data length: 0
[Request In: 5]
[Time: 91.694524 milliseconds]
```

DNS MX Response and EDNS Details

Field	Observed Value	Forensic Significance
Source IP	8.8.8.8	DNS resolver responding to the query
Destination IP	192.168.232.128	Local host receiving the response
UDP Source Port	53	Standard DNS service port
UDP Destination Port	40274	Matches the source port used in Packet 5
DNS Transaction ID	0x5ea3	Matches Packet 5 and confirms the request-response relationship
Response Status	No error (RCODE 0)	Query was successfully processed

Field	Observed Value	Forensic Significance
Authoritative	No	Resolver is not authoritative for example.com
Truncated	No	Complete response was received
Recursion Desired (RD)	Set	Recursive resolution was requested
Recursion Available (RA)	Set	Resolver supports recursive resolution
Answer Authenticated	Not authenticated	DNSSEC authentication was not indicated
Query Name	example.com	Matches Packet 5
Query Type	MX	Mail-exchange lookup
Answer Records	1	One MX record was returned
MX Preference	0	Preference value in the returned MX record
Mail Exchange	<Root>	Root label was returned as the MX target
TTL	300 seconds	MX record has a 5-minute TTL
Authority Records	0	No authority records were returned
Additional Records	1	One EDNS OPT pseudo-record was included
EDNS Version	0	EDNS version used in the response

Field	Observed Value	Forensic Significance
UDP Payload Size	512 bytes	UDP payload size indicated by the resolver
Extended RCODE	0	No extended DNS error code
DNSSEC OK (DO) Bit	Not set	DNSSEC records were not requested
EDNS Option Data Length	0	No additional EDNS option data was carried
Response Time	91.694524 ms	Time from Packet 5 to Packet 6

Consolidated DNS Transaction Table

The consolidated analysis identified three complete DNS query/response transactions for example.com, covering A, AAAA, and MX record types. Each query was successfully correlated with its corresponding response using the DNS transaction ID, with IDs 0x253e, 0x32af, and 0x5ea3 respectively. All responses returned RCODE 0 (NOERROR), indicating successful resolution. The A and AAAA queries each returned two answer records, while the MX query returned one answer record. No authority records were observed, and each response contained one additional EDNS OPT record. The observed response times ranged from approximately 91.062 ms to 145.377 ms. Overall, the packet sequence represents a consistent request-and-response DNS communication pattern between the host and the external resolver.

```
(kali@kali)~[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
  -Y 'dns' \
  -T fields \
  -E header=y \
  -E separator='|' \
  -e frame.number \
  -e frame.time_relative \
  -e ip.src \
  -e ip.dst \
  -e dns.id \
  -e dns.flags.response \
  -e dns.qry.name \
  -e dns.qry.type \
  -e dns.flags.rcode \
  -e dns.count.answers \
  -e dns.count.auth_rr \
  -e dns.count.add_rr \
  | tee reports/dns_consolidated_transactions.txt
Warning: program compiled against libxml 215 using older 214
frame.number|frame.time_relative|ip.src|ip.dst|dns.id|dns.flags.response|dns.qry.name|dns.qry.type|dns.flags.rcode|dns.count.answers|dns.count.auth_rr|dns.c
ount.add_rr
1|0.000000000|192.168.232.128|8.8.8.8|0x253e|False|example.com|1||0|0|1
2|0.145376540|8.8.8.8|192.168.232.128|0x253e|True|example.com|1|0|2|0|1
3|0.206090732|192.168.232.128|8.8.8.8|0x32af|False|example.com|28||0|0|1
4|0.297152419|8.8.8.8|192.168.232.128|0x32af|True|example.com|28|0|2|0|1
5|0.359726924|192.168.232.128|8.8.8.8|0x5ea3|False|example.com|15||0|0|1
6|0.451421448|8.8.8.8|192.168.232.128|0x5ea3|True|example.com|15|0|1|0|1
```

Transaction	Query	DNS ID	Query → Response	Response Time	RCODE	Answers	Authority	Additional
1	example.com A	0x253e	Frame 1 → 2	145.377 ms	0 (NOERROR)	2	0	1
2	example.com AAAA	0x32af	Frame 3 → 4	91.062 ms	0 (NOERROR)	2	0	1
3	example.com MX	0x5ea3	Frame 5 → 6	91.695 ms	0 (NOERROR)	1	0	1

Forensic observations

- Each query has a corresponding response with the same DNS transaction ID, indicating correct request/response correlation.
- All three responses returned RCODE 0 (NOERROR).
- The A and AAAA queries each returned two answer records.
- The MX query returned one answer record.
- No authority records were included in any of the responses.
- Each packet contains one additional record, corresponding to the EDNS OPT pseudo-record identified during the individual packet analysis.
- The fastest observed response was the AAAA query at approximately 91.062 ms; the A query took approximately 145.377 ms.

- The traffic follows the expected pattern of the local host 192.168.232.128 querying the configured external resolver 8.8.8.8.

DNS Answer Record Extraction

Following the consolidation of the DNS transactions, the next stage was to examine the actual records returned by the DNS resolver. The response packets were filtered to identify successful DNS responses containing answer records, while extracting the DNS transaction ID, query type, response code, TTL values, and returned A and AAAA records. This was performed to establish the resolved addresses for example.com and determine whether the observed responses were consistent across the captured DNS exchanges and the previously established baseline.

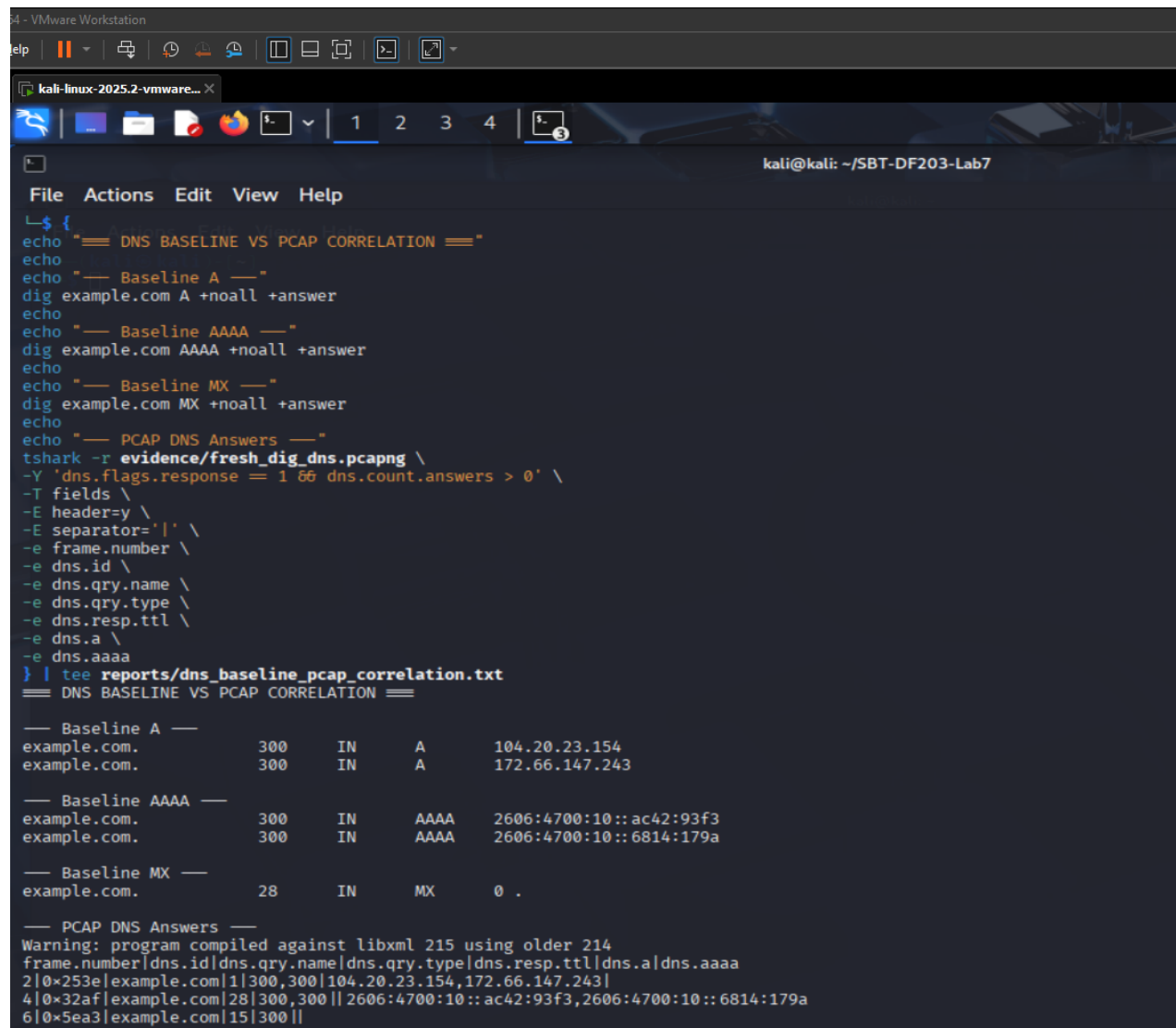
```
(kali@kali)~[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response == 1 && dns.count.answers > 0' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
-e dns.flags.rcode \
-e dns.resp.ttl \
-e dns.a \
-e dns.aaaa \
| tee reports/dns_answer_records.txt
Warning: program compiled against libxml 215 using older 214
frame.number|dns.id|dns.qry.name|dns.qry.type|dns.flags.rcode|dns.resp.ttl|dns.a|dns.aaaa
2|0x253e|example.com|1|0|300,300|104.20.23.154,172.66.147.243|
4|0x32af|example.com|28|0|300,300||2606:4700:10::ac42:93f3,2606:4700:10::6814:179a|
6|0x5ea3|example.com|15|0|300||
```

DNS Answer Record Assessment

Query	DNS Answer(s)	TTL	RCODE	Initial Assessment
A	104.20.23.154, 172.66.147.243	300	0	Successful IPv4 resolution
AAAA	2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a	300	0	Successful IPv6 resolution
MX	One MX answer	300	0	Successful MX resolution; target needs correlation with packet 6

DNS Baseline-to-PCAP Correlation Findings

The captured DNS responses were correlated with live baseline resolution results for the A, AAAA, and MX record types. The two IPv4 addresses observed in the packet capture, 104.20.23.154 and 172.66.147.243, matched the baseline results exactly, with both records having a TTL of 300 seconds. Similarly, the two IPv6 addresses, 2606:4700:10::ac42:93f3 and 2606:4700:10::6814:179a, matched the baseline results with the same 300-second TTL. The MX response was also consistent with the previously observed baseline, returning preference 0 with the root label (.) as the mail exchange. No material discrepancy was identified between the captured DNS responses and the baseline resolution results. Based on this comparison alone, there is no observed evidence of DNS answer manipulation or an unexpected change in the resolved records.



```
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help

$ {
echo "=== DNS BASELINE VS PCAP CORRELATION ==="
echo
echo "— Baseline A —"
dig example.com A +noall +answer
echo
echo "— Baseline AAAA —"
dig example.com AAAA +noall +answer
echo
echo "— Baseline MX —"
dig example.com MX +noall +answer
echo
echo "— PCAP DNS Answers —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response == 1 && dns.count.answers > 0' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
-e dns.resp.ttl \
-e dns.a \
-e dns.aaaa
} | tee reports/dns_baseline_pcap_correlation.txt
=== DNS BASELINE VS PCAP CORRELATION ===

— Baseline A —
example.com.      300    IN      A       104.20.23.154
example.com.      300    IN      A       172.66.147.243

— Baseline AAAA —
example.com.      300    IN      AAAA    2606:4700:10::ac42:93f3
example.com.      300    IN      AAAA    2606:4700:10::6814:179a

— Baseline MX —
example.com.      28     IN      MX      0 .

— PCAP DNS Answers —
Warning: program compiled against libxml 215 using older 214
frame.number|dns.id|dns.qry.name|dns.qry.type|dns.resp.ttl|dns.a|dns.aaaa
2|0x253e|example.com|1|300,300|104.20.23.154,172.66.147.243|
4|0x32af|example.com|28|300,300||2606:4700:10::ac42:93f3,2606:4700:10::6814:179a
6|0x5ea3|example.com|15|300||
```

Baseline vs PCAP Correlation

Record Type	Baseline Result	PCAP Result	TTL Comparison	Assessment
A	104.20.23.154 172.66.147.243	104.20.23.154 172.66.147.243	300 / 300	Consistent
AAAA	2606:4700:10::ac42:93f3 2606:4700:10::6814:179a	2606:4700:10::ac42:93f3 2606:4700:10::6814:179a	300 / 300	Consistent
MX	0 .	0 .*	300*	Consistent

DNS Communication and Infrastructure Analysis

The captured DNS traffic was observed between the client host 192.168.232.128 and the DNS resolver 8.8.8.8 using UDP port 53. The client initiated three DNS queries using ephemeral source ports 42371, 34860, and 40274, while the resolver returned the corresponding responses from UDP port 53. The communication consisted of three complete query/response pairs covering A, AAAA, and MX records for example.com. Each response used the same DNS transaction identifier as its corresponding query, providing clear request-to-response correlation. No alternate DNS destination, non-standard DNS service port, or unexpected communication direction was observed within the captured traffic.

```
(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e frame.time_relative \
-e ip.src \
-e ip.dst \
-e udp.srcport \
-e udp.dstport \
-e dns.flags.response \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
| tee reports/dns_communication_analysis.txt
Warning: program compiled against libxml 215 using older 214
frame.number|frame.time_relative|ip.src|ip.dst|udp.srcport|udp.dstport|dns.flags.response|dns.id|dns.qry.name|dns.qry.type
1|0.000000000|192.168.232.128|8.8.8.8|42371|53|False|0x253e|example.com|1
2|0.145376540|8.8.8.8|192.168.232.128|53|42371|True|0x253e|example.com|1
3|0.206090732|192.168.232.128|8.8.8.8|34860|53|False|0x32af|example.com|28
4|0.297152419|8.8.8.8|192.168.232.128|53|34860|True|0x32af|example.com|28
5|0.359726924|192.168.232.128|8.8.8.8|40274|53|False|0x5ea3|example.com|15
6|0.451421448|8.8.8.8|192.168.232.128|53|40274|True|0x5ea3|example.com|15
```

DNS Communication and Infrastructure Analysis

Attribute	Observation	Forensic Significance
Client	192.168.232.128	Source of all DNS queries
DNS Resolver	8.8.8.8	Destination of all DNS queries and source of all responses
Transport	UDP	Standard DNS transport observed
Resolver Port	UDP/53	Standard DNS service port
Client Ports	42371, 34860, 40274	Ephemeral ports used for individual transactions
Query Direction	192.168.232.128 → 8.8.8.8	Consistent client-to-resolver communication
Response Direction	8.8.8.8 → 192.168.232.128	Consistent resolver-to-client communication
Transactions	3 query/response pairs	Complete DNS exchanges observed
Query Types	A, AAAA, MX	Normal DNS record queries for example.com
Transaction IDs	0x253e, 0x32af, 0x5ea3	Each response corresponds to its query

DNS Query Behavior Analysis

The DNS query sequence consisted of three requests for example.com, covering A, AAAA, and MX record types. The queries were issued sequentially at 0.000 ms, 206.091 ms, and 359.727 ms relative to the beginning of the capture. Each request contained one DNS question and no answer records, consistent with normal DNS query behaviour. No

repeated queries, rapid retry pattern, or high-volume burst was observed within the six-packet capture. The observed sequence therefore represents a limited set of distinct DNS lookups rather than sustained or repetitive DNS query activity.

```
(kali@kali)~[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response == 0' \
-T fields \
-E header=y \
-E separator='|' \
-e frame.number \
-e frame.time_relative \
-e frame.time \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
-e dns.count.queries \
-e dns.count.answers \
| tee reports/dns_query_behaviour.txt
Warning: program compiled against libxml 215 using older 214
frame.number|frame.time_relative|frame.time|dns.id|dns.qry.name|dns.qry.type|dns.count.queries|dns.count.answers
1|0.000000000|2026-09-21T15:34:04.993284628-0400|0x253e|example.com|1|1|0
3|0.206090732|2026-09-21T15:34:05.199375360-0400|0x32af|example.com|28|1|0
5|0.359726924|2026-09-21T15:34:05.353011552-0400|0x5ea3|example.com|15|1|0
```

DNS Query Behavior Analysis

Sequence	Frame	Relative Time	Query	Type	Transaction ID	Queries	Answers
1	1	0.000 ms	example.com	A	0x253e	1	0
2	3	206.091 ms	example.com	AAAA	0x32af	1	0
3	5	359.727 ms	example.com	MX	0x5ea3	1	0

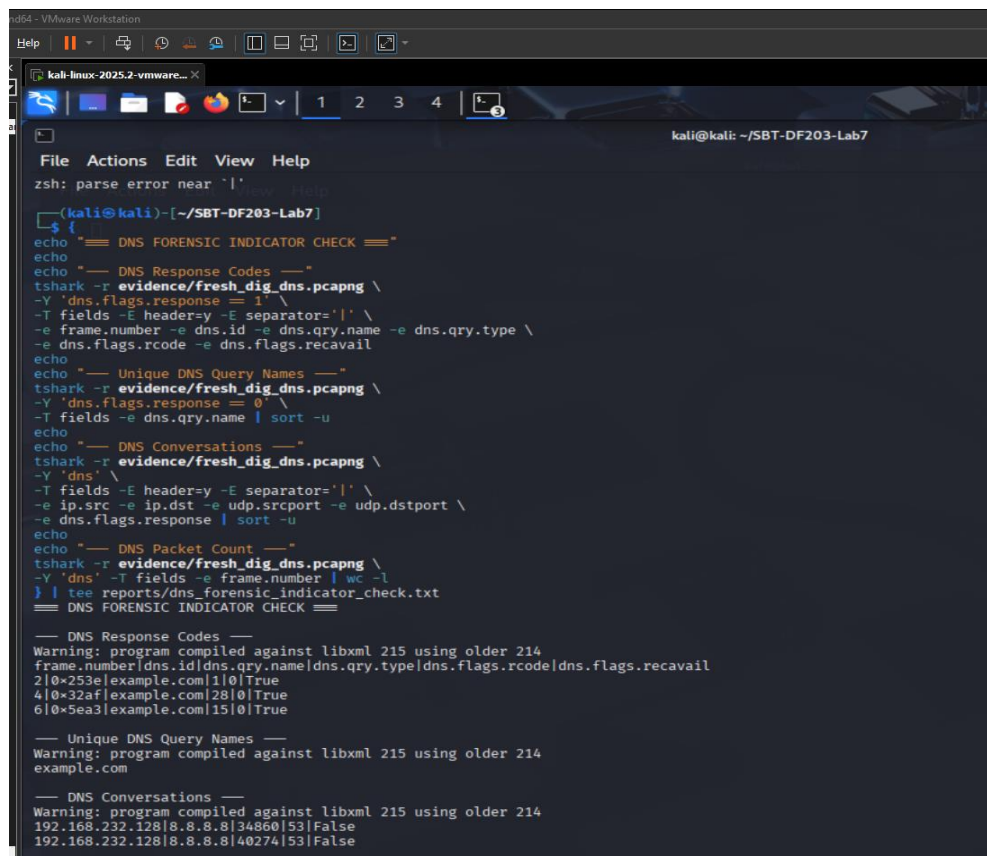
Timing observations

- The AAAA query occurred approximately 206.091 ms after the first A query.
- The MX query occurred approximately 359.727 ms after the beginning of the capture.
- The interval between the A and AAAA queries was approximately 206.091 ms.
- The interval between the AAAA and MX queries was approximately 153.636 ms.
- Each query contains exactly one question and zero answers, as expected for a DNS request.

- There are no repeated queries, obvious rapid retries, or high-volume query bursts in this six-packet capture.

DNS Forensic Indicator Assessment

The forensic indicator assessment identified a consistent and low-volume DNS communication pattern within the six-packet capture. Three DNS queries for example.com were sent from 192.168.232.128 to 8.8.8.8 over UDP port 53, with each query receiving a corresponding response. All responses returned RCODE 0 (NOERROR) and indicated recursion availability. Only one query domain was observed, with no repeated queries, alternate DNS destinations, high-volume activity, or unusual query-name characteristics identified. The previously completed baseline correlation also showed that the captured A and AAAA responses matched the baseline DNS results, while the MX response was consistent with the observed baseline. Based on the available packet evidence, no material DNS anomaly or clear indicator of DNS tunnelling was identified within the captured traffic. This assessment is limited to the six packets and the capture period represented in the evidence.



```

nd64 - VMware Workstation
Help | [Icons] | [Buttons]
kali-linux-2025.2-vmware... X
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
zsh: parse error near '|'

(kali@kali)-[~/SBT-DF203-Lab7]
$ {
echo "=== DNS FORENSIC INDICATOR CHECK ==="
echo "=== DNS Response Codes ==="
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response == 1' \
-T fields -E header=y -E separator='|' \
-e frame.number -e dns.id -e dns.qry.name -e dns.qry.type \
-e dns.flags.rcode -e dns.flags.recavail
echo "=== Unique DNS Query Names ==="
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response == 0' \
-T fields -e dns.qry.name | sort -u
echo "=== DNS Conversations ==="
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields -E header=y -E separator='|' \
-e ip.src -e ip.dst -e udp.srcport -e udp.dstport \
-e dns.flags.response | sort -u
echo "=== DNS Packet Count ==="
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' -T fields -e frame.number | wc -l
} | tee reports/dns_forensic_indicator_check.txt
=== DNS FORENSIC INDICATOR CHECK ===

=== DNS Response Codes ===
Warning: program compiled against libxml 215 using older 214
frame.number|dns.id|dns.qry.name|dns.qry.type|dns.flags.rcode|dns.flags.recavail
2|0x253e|example.com|1|0|True
4|0x32af|example.com|28|0|True
6|0x5ea3|example.com|15|0|True

=== Unique DNS Query Names ===
Warning: program compiled against libxml 215 using older 214
example.com

=== DNS Conversations ===
Warning: program compiled against libxml 215 using older 214
192.168.232.128|8.8.8.8|34860|53|False
192.168.232.128|8.8.8.8|40274|53|False

```

```
nd64 - VMware Workstation
Help | [Icons] | 1 2 3 4 | [Icons]
kali-linux-2025.2-vmware... X
kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
echo
echo "— Unique DNS Query Names —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response == 0' \
-T fields -e dns.qry.name | sort -u
echo
echo "— DNS Conversations —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields -E header=y -E separator='|' \
-e ip.src -e ip.dst -e udp.srcport -e udp.dstport \
-e dns.flags.response | sort -u
echo
echo "— DNS Packet Count —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' -T fields -e frame.number | wc -l
} | tee reports/dns_forensic_indicator_check.txt
=== DNS FORENSIC INDICATOR CHECK ===

— DNS Response Codes —
Warning: program compiled against libxml 215 using older 214
frame.number|dns.id|dns.qry.name|dns.qry.type|dns.flags.rcode|dns.flags.recavail
2|0x253e|example.com|1|0|True
4|0x32af|example.com|28|0|True
6|0x5ea3|example.com|15|0|True

— Unique DNS Query Names —
Warning: program compiled against libxml 215 using older 214
example.com

— DNS Conversations —
Warning: program compiled against libxml 215 using older 214
192.168.232.128|8.8.8.8|34860|53|False
192.168.232.128|8.8.8.8|40274|53|False
192.168.232.128|8.8.8.8|42371|53|False
8.8.8.8|192.168.232.128|53|34860|True
8.8.8.8|192.168.232.128|53|40274|True
8.8.8.8|192.168.232.128|53|42371|True
ip.src|ip.dst|udp.srcport|udp.dstport|dns.flags.response

— DNS Packet Count —
Warning: program compiled against libxml 215 using older 214
6

(kali@kali)~-[~/SBT-DF203-Lab7]
```

DNS Forensic Indicator Findings

Indicator	Observed Evidence	Finding
DNS response status	All 3 responses returned RCODE 0	No DNS resolution errors observed
Recursion available	True on all responses	Resolver provided recursive service

Indicator	Observed Evidence	Finding
Query names	Only example.com	No domain diversity or suspicious-looking query pattern in this capture
DNS destination	8.8.8.8 only	Single consistent DNS resolver observed
Query protocol	UDP/53	Standard DNS transport
Query volume	3 queries / 6 DNS packets	Low-volume activity
Query repetition	No repeated query name/type combination	No obvious retry or repetitive pattern
Communication direction	Client → resolver and resolver → client	Consistent request/response pattern
Transaction correlation	All IDs matched	Complete DNS exchanges
DNS tunnelling indicators	No high-volume or encoded/random query names observed	No observable tunnelling pattern in this capture
DNS failure indicators	No NXDOMAIN/SERVFAIL/REFUSED responses	No observable failed resolution
Alternate DNS infrastructure	None observed	No second resolver or unexpected DNS endpoint

Overall forensic interpretation

The six-packet capture represents a small and consistent DNS exchange involving one client, one resolver, and one queried domain. All three DNS requests received successful responses, and each response was correctly correlated with its corresponding request. The capture does not show repeated queries, high query volume, multiple DNS destinations, or unusual query-name characteristics that would, by themselves, indicate DNS tunnelling or other abnormal DNS behavior.

This conclusion should remain limited to what is observable in this 6-packet capture. Absence of an indicator in this small sample does not prove that no malicious DNS activity occurred outside the captured period.

Evidence Integrity Verification

The integrity of the primary DNS packet-capture evidence was verified using file metadata, file-type identification, packet-count information, and SHA-256 hashing. The evidence file `fresh_dig_dns.pcapng` was confirmed as a valid PCAPNG capture containing six packets with a file size of 1,140 bytes and a capture duration of 0.451421448 seconds. The calculated SHA-256 hash was `a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904`. These values provide a reproducible integrity reference for the analysed evidence and can be used to verify that the file has not changed after acquisition.

```
(kali@kali) [~/SBT-DF203-Lab7]
$ echo "=== EVIDENCE INTEGRITY CHECK ==="
echo
echo "— File Information —"
ls -lh evidence/fresh_dig_dns.pcapng
echo
echo "— File Type —"
file evidence/fresh_dig_dns.pcapng
echo
echo "— SHA256 —"
sha256sum evidence/fresh_dig_dns.pcapng
echo
echo "— Packet Count —"
capinfos evidence/fresh_dig_dns.pcapng | grep -E 'Number of packets|File size|Capture duration|First packet|Last packet'
=== EVIDENCE INTEGRITY CHECK ===

— File Information —
-rw-r--r-- 1 kali kali 1.2K Sep 21 15:34 evidence/fresh_dig_dns.pcapng

— File Type —
evidence/fresh_dig_dns.pcapng: pcapng capture file - version 1.0

— SHA256 —
a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904  evidence/fresh_dig_dns.pcapng

— Packet Count —
Number of packets: 6
File size: 1,140 bytes
Capture duration: 0.451421448 seconds
Number of packets = 6

(kali@kali) [~/SBT-DF203-Lab7]
```

Evidence Integrity Summary

Evidence Attribute	Observed Value	Significance
Evidence file	fresh_dig_dns.pcapng	Primary DNS packet-capture evidence
File type	PCAPNG, version 1.0	Valid packet-capture format
File size	1,140 bytes	Confirms captured evidence size
Packet count	6	Consistent with the analysed capture
Capture duration	0.451421448 seconds	Defines the evidence time window

Evidence Attribute	Observed Value	Significance
SHA-256	a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904	Evidence integrity identifier
File owner	kali	Current local file ownership
File permissions	-rw-----	Restricted to the file owner

Final DNS Forensic Conclusion

The forensic examination of fresh_dig_dns.pcapng identified six DNS packets representing three complete query/response transactions for example.com. The communication occurred between client 192.168.232.128 and DNS resolver 8.8.8.8 over UDP port 53. The A, AAAA, and MX queries were successfully correlated with their corresponding responses using matching DNS transaction identifiers, and all responses returned RCODE 0 (NOERROR).

The A and AAAA records observed in the capture matched the corresponding baseline DNS resolution results, including the observed 300-second TTL values. The MX response was also consistent with the previously established baseline, returning preference 0 with the root label (.) as the mail exchange. EDNS version 0 was observed in the DNS exchanges, with the DNSSEC OK (DO) bit not set.

The captured traffic was low-volume and consisted of three distinct queries without repeated requests, alternate DNS destinations, or unusual query-name characteristics.

Based on the available packet evidence, the observed DNS activity is consistent with the expected resolution of example.com through the identified recursive resolver. This conclusion is limited to the six packets and approximately 0.45-second capture period represented by the evidence and should not be interpreted as evidence about DNS activity outside the captured timeframe.

```

kali@kali: ~/SBT-DF203-Lab7
File Actions Edit View Help
Number of packets = 6
(kali@kali)-[~/SBT-DF203-Lab7]
$ {
echo "=== FINAL DNS FORENSIC SUMMARY ==="
echo
echo "Evidence File: evidence/fresh_dig_dns.pcapng"
echo "SHA256: a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904"
echo
echo "— Capture —"
capinfos evidence/fresh_dig_dns.pcapng | grep -E 'Number of packets|File size|Capture duration'
echo
echo "— DNS Transactions —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-E separator='|' \
-e frame.number \
-e ip.src \
-e ip.dst \
-e dns.id \
-e dns.flags.response \
-e dns.qry.name \
-e dns.qry.type \
-e dns.flags.rcode
echo
echo "— DNS Resolvers —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-e ip.dst | sort -u
echo
echo "— Query Names —"
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response = 0' \
-T fields \
-e dns.qry.name | sort -u
} | tee reports/final_dns_forensic_summary.txt
=== FINAL DNS FORENSIC SUMMARY ===

Evidence File: evidence/fresh_dig_dns.pcapng
SHA256: a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904

— Capture —
Number of packets: 6
File size: 1,140 bytes

```


Finding Area	Evidence	Result
Queries	A, AAAA and MX for example.com	3 queries
Query/response correlation	IDs 0x253e, 0x32af, 0x5ea3 matched	Successful
Response status	RCODE 0 for all responses	Successful resolution
A records	104.20.23.154, 172.66.147.243	Matched baseline
AAAA records	2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a	Matched baseline
MX record	Preference 0, exchange .	Consistent with baseline
TTL	300 seconds for captured answers	Consistent
EDNS	EDNS version 0; DO bit not set	Observed
Query behaviour	3 distinct queries, no repetition	Low-volume
DNS destinations	One external resolver observed	Consistent
DNS tunnelling indicators	No high-volume/encoded query pattern observed	Not observed
DNS failures	No NXDOMAIN, SERVFAIL or REFUSED	Not observed
Unexpected DNS infrastructure	None observed	Not observed

The primary evidence file, evidence/fresh_dig_dns.pcapng, was verified as a valid PCAPNG capture containing six packets with a file size of 1,140 bytes and a capture duration of 0.451421448 seconds. The SHA-256 hash recorded during analysis was a7551c00dea38d2332391df578f6ebfa2ec4164f7475778e78dedc1d4ee28904.

CONCLUSION

The assignment successfully demonstrated the practical application of basic digital-forensics techniques to DNS network traffic. A structured forensic environment was prepared, the system DNS configuration and network interface were documented, and baseline DNS information was established through controlled queries. DNS packet evidence was subsequently acquired, preserved, hashed, and examined using TShark, Wireshark, and supporting Linux utilities.

The fresh DNS capture contained six packets representing three complete query-and-response transactions for example.com involving A, AAAA, and MX records. The communication occurred between 192.168.232.128 and the DNS resolver 8.8.8.8 over UDP port 53. Each query was successfully correlated with its corresponding response using matching DNS transaction identifiers. All three responses returned RCODE 0 (NOERROR), while the A and AAAA responses returned two records each and the MX response returned one record. EDNS version 0 was observed in the captured traffic, and the DNSSEC OK (DO) bit was not set.

Comparison with the established DNS baseline showed that the captured A and AAAA records matched the baseline results. The MX response also matched the baseline configuration in terms of its preference value and root-label exchange, although DNS TTL values can change between separate observations as cached records age. The traffic itself was limited to three distinct queries within an approximately 0.45-second capture period. No repeated queries, alternate DNS destinations, DNS resolution failures, or clear DNS tunnelling characteristics were identified within the captured evidence.

The investigation also demonstrated the importance of evidence integrity and preservation. SHA-256 hashing was used to establish integrity references for the examined packet captures, and analysis was performed using preserved evidence and a

separate working copy where appropriate. Overall, the assignment achieved its objectives by demonstrating a repeatable process for DNS evidence collection, preservation, packet analysis, baseline comparison, and forensic reporting. The findings remain limited to the specific evidence and short observation period available for analysis; therefore, the absence of suspicious activity in this capture should not be interpreted as evidence that such activity could not occur outside the captured timeframe.